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Entity and Relation Extraction from Electronic Health Records using SpaCy

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Summary

En esta investigación adaptamos un método innovador para extraer entidades y relaciones de historias clínicas (EHRs) en el ámbito médico. Realizamos pruebas en vasco, italiano y español para observar el comportamiento de la aproximación en un contexto multilingüe. La ventaja de este método es la facilidad para cambiar el modelo del lenguaje base. Su reusabilidad reside en que solo requiere actualizar los datos de entrenamiento para las tareas de Reconocimiento de Entidades Nombradas (NER) y Extracción de Relaciones (RE).

Abstract

In this research we adapt an innovative method to extract entities and relationships from medical records (EHRs) in the medical domain. We performed tests in Basque, Italian and Spanish to observe the behaviour of the approach in a multilingual context. The advantage of this method is the ease of changing the base language model. Its reusability resides in the fact that it only requires updating the training data for the Named Entity Recognition (NER) and Relationship Extraction (RE) tasks.

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Capítulo 1

Introduction

Artificial Intelligence (AI) refers to the process of simulating human-like intelligence to machines. It has significantly impacted many fields, including **Natural Language Processing (NLP)**.

NLP has been considered a subfield of AI [1] from the beginning, whose main focus is on analyzing and understanding human language. NLP involves many tasks, including machine translation, sentiment analysis, text summarization, and speech recognition, amongst others. AI seems to offer the necessary instruments and approaches to comprehend human language, which is an elaborate and complex mode of communication.

The development of deep learning techniques in the past years has had a significant impact on the field of NLP [2]. Advanced deep learning models, such as Recurrent Neuronal Networks (RNNs), Convolutional Neuronal Networks (CNNs), and Transformer-based models, have demonstrated outstanding results in several NLP tasks. These models can learn and deduce significant patterns from extensive datasets, which enables them to manage some of the difficulties of natural language.

Moreover, recent developments in AI and NLP have also led to the emergence of new applications, such as language models that can generate natural language text and can be used to respond to human queries conversationally, known as conversational AI or chatbots [3]. These applications have shown promising results in improving customer experience and reducing the workload of customer support teams.

Furthermore, the surge in transformer-based models [4] has enhanced the extraction of complex patterns and relationships, which are essential for tasks like event extraction and argument mining. However, challenges such as ambiguity among phrases, implicit knowledge, and lack of contextual understanding promoted by the domain of the text still exist.

Regarding **Information Extraction (IE)** field, where significant advances have been made with language models, in this work, we will fundamentally look at two crucial tasks within IE, namely, Named Entity Recognition (NER) and Relation Extraction (RE) at the front of this evolution in the medical domain [5].

Named Entity Recognition (NER) [6] is a fundamental NLP task that involves identifying and classifying named entities in text into predefined categories such as a

person, organization, location, date, and so on. Named entities are words or phrases (N-grams) that refer to specific entities in the real world, such as people, places, organizations, and products.

There are various approaches to performing NER, including rule-based and machine learning-based methods under supervised and unsupervised views.

Rule-based methods are based on pre-defined rules or patterns to identify named entities in text. For example, a rule-based system may search for specific words or patterns to identify the name of a person or organization. Rule-based methods are often simple, fast, and explainable, but may lack the flexibility and robustness to be applied in different domains, contrasting with machine learning-based methods.

Machine learning-based methods involve training models on labelled data to predict the category of each word or phrase in the text. These models can be supervised or unsupervised and can use various features such as part-of-speech tags, dependency trees, and word embeddings to represent the text. Deep learning models, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), among others, have achieved state-of-the-art performance on NER tasks.

One of the challenges of NER is the ambiguity problem in natural language, which can make it difficult to identify the correct category for a named entity. For example, the word 'apple' can refer to a fruit or a company, and the word 'Paris' can refer to a city or a person's name. Another challenge is the variability in the expression of named entities in text, such as abbreviations, misspellings, and multiple-word expressions.

As for **Relation Extraction (RE)** [7], it involves identifying and extracting the relationships between entities in text. This task is essential in many NLP applications too, such as information extraction, knowledge graph construction, question answering, and chatbots.

However, this is also a challenging task, much like NER, particularly when dealing with noisy data and unstructured text. Such data can include text from social media or unstructured web content, which often presents considerable variability in the expression of relationships. In short, text from different persons is written in different ways. The same information or meaning can be conveyed by different natural language expressions.

Entities can be referred to by using different names or pronouns, and relationships can be expressed using synonymous expressions or paraphrases. Another challenge is the ambiguity in natural language, which can make it difficult to identify the correct relationship between entities.

To overcome these challenges, researchers have developed a variety of techniques and tools that combine machine learning, deep learning, and linguistic knowledge.

In the text, relationships between entities can be expressed in various ways, including verbs, prepositions or adjectives, or even none of them (implicit relationship). For instance, in the sentence 'Steve Jobs co-founded Apple', the entities would be 'Steve Jobs' and 'Apple' as nouns, and the relationship between them would be 'co-founded' as a verb.

When it comes to relation extraction, there are several methods and techniques available to identify and extract relationships between entities in text, as was the case

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with NER; rule-based or machine learning-based methods.

Rule-based methods rely on pre-defined rules or patterns to identify relationships between entities in text. For example, a rule-based system might search for the verb 'is married to' to identify the relationship between two entities in a sentence, such as 'Barack Obama is married to Michelle Obama'. However, it still has the same problem of use. As the example shows, it only works for specific words or patterns, in this case 'is married to'.

Machine Learning methods leverage computational models to learn and identify relationships between entities in text. For example, a supervised learning model could be trained on a dataset in which relationships are already annotated, such as the sentence 'Barack Obama is married to Michelle Obama.' Here, the model learns from the labelled relationship 'is married to' between the entities 'Barack Obama' and 'Michelle Obama'. Over time, these models can generalize their learning to accurately identify relationships in a new unseen text.

Despite the significant role of rule-based and traditional machine learning methods for NER and RE, the focus of recent work has predominantly been on transformer-based models. These models, introduced in the paper 'Attention is All You Need' [4], mentioned above, leverage deep learning methodologies and often yield more profound and complex insights compared to standard machine learning approaches.

The 'attention' mechanism they employ weighs the importance of different words in the text when determining the context, enabling them to outperform previous methods in tasks like NER and RE.

Popular transformer-based models like BERT, GPT, and RoBERTa have significantly contributed to the recent surge in the field. They have notably improved the extraction of complex patterns and relationships, which are crucial for more applications such as event extraction and argument mining. However, despite their success, challenges persist, particularly influenced by the domain of the text.

In the case of NER, a transformer model can effectively understand the context of a sentence and, therefore, better identify and classify entities. For RE, these models can discern complex relationships between entities, even when they are not explicitly stated, thanks to their ability to capture deep semantic relationships in the text.

1.1. Motivation

The work documented in this Master's Thesis aims at contributing to the information extraction task from Electronic Health Records (EHRs), a crucial source of patient information. In this context, the recognition of entities and the relations between them can help to improve the decision-making process. Existing medical NER models identify specific biological entities like symptoms, diseases, tests, and results from text data [8, 9], while medical RE models discern relationships between such entities within the extensive medical literature [10, 11].

Electronic Health Records (EHRs) are digital versions of a patient's medical history, which are maintained by the provider (private hospitals or public health care) over time. EHRs may include all of the key administrative clinical data relevant to that person's care, including demographics, progress notes, problems, medications, vital

signs, past medical history, immunizations, laboratory data, and radiology reports. The EHR automates access to information and has the potential to streamline the clinician’s workflow. EHRs are designed to be accessed by all individuals involved in the patient’s care, including the patients themselves.

In the context of this study, our primary focus will be on various types of medical data such as progress notes, problems, medications, vital signs, past medical history, immunizations, and laboratory data, preferently, we are interested in the content of a doctor’s notes during a consultation.

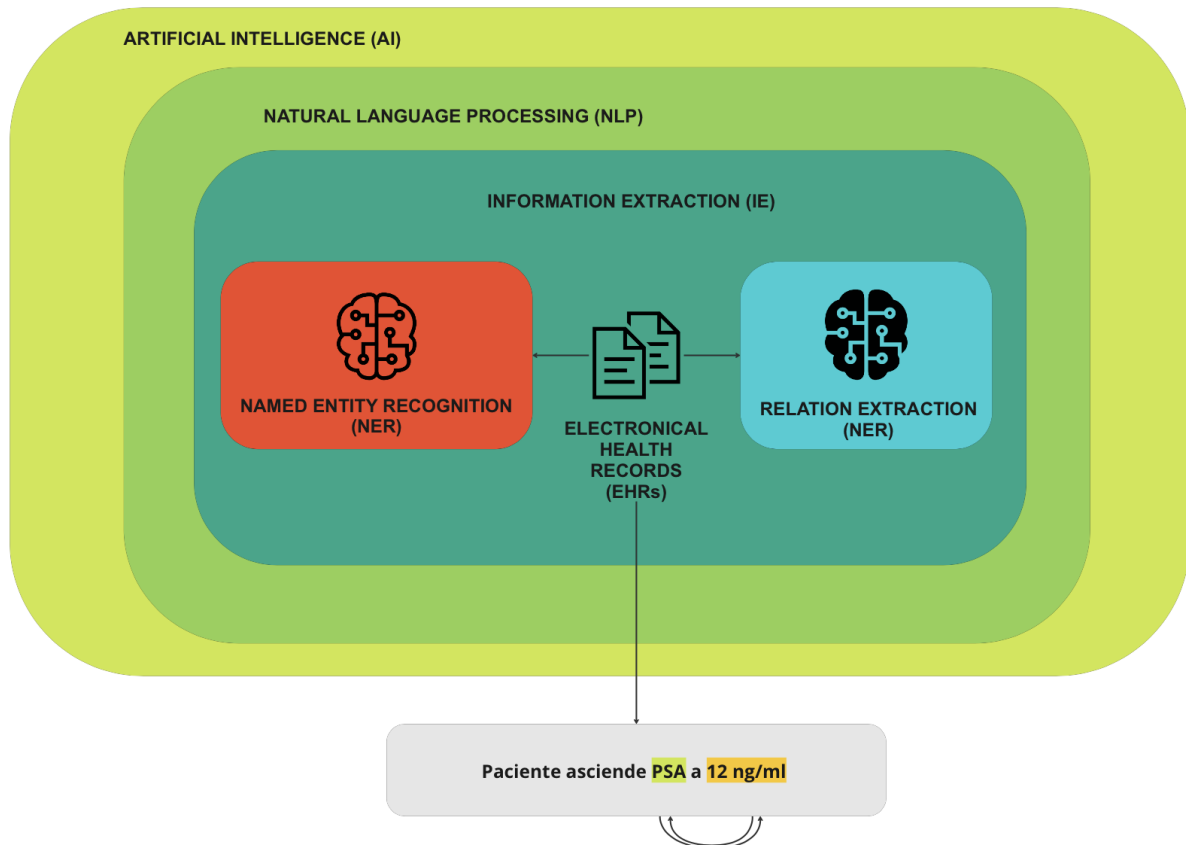


Figura 1.1: Preprocessing workflow.

Figure 1.1 presents the structure of this work, illustrating the utilization of NER and RE in processing EHRs for information extraction purposes.

Driven by these considerations, we chose to participate in a shared evaluation campaign of Natural Language Processing systems conducted, mainly, in Spanish. We put forward a solution for a RE task as part of the campaign TESTLINK@IberLEF 2023 - evaluation in Spanish and Basque- and CLinkART@Evalita 2023 - evaluation in Italian. The task involved extracting relationships from clinical cases featured in the E3C corpus. This corpus comprises Spanish, Italian, and Basque written documents that detail aspects of clinical practice, such as the reasons for a clinical visit, physical examinations conducted, patient diagnosis evaluations, and subsequent treatments. Further details can be found to the hyperlinks provided in the footnotes¹. These tasks consist in identifying test results and measurements and linking them

¹<https://e3c.fbk.eu/testlinkiberlef> and <https://e3c.fbk.eu/clinkart>

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to the textual mentions of the laboratory tests and measurements from which they were obtained. Given the context of these languages (Spanish, Basque, and Italian), this work is positioned within the scope of Natural Language Processing (NLP) with limited resources.

By these datasets, our objective is to detect and extract entities that are characterized as EVENTS and RMLs (Results, Measurements and Laboratories). To provide an example, within the outcomes of an blood analysis translated to the doctor's notes, the elements identified would be classified as EVENTS, while the corresponding numerical values would be RMLs. Not all values of RMLs are numerical, they could be categorical as well.

Additionally, we will also extract the relations between these entities to gain further insight into the semantics of the medical data. This approach is designed to yield a robust understanding of the complex role of EVENTS and RMLs within a healthcare context.

However, there are several obstacles that hinder IE tasks from health records. The first one is that both NER and RE models need to be trained with abundant examples in the language(s) object of analysis. The second obstacle refers to the fact that those corpora have to be annotated by experts in the domain who should have a good command of the languages involved, which is a time-consuming task, prone to errors, and often not even feasible.

The main motivation for this work comes from the observation that current solutions to many of these tasks are predominantly in English, thus distancing other languages from these advancements. One of the key challenges faced is finding appropriate datasets, even in English. Ideally, these datasets should be precisely annotated and available in several languages. The majority of such datasets require the skills of language experts or annotators, specialized in the specific domain for precise annotation. Such a dataset often comes from a specific field or domain, such as biomedical literature or news articles, and it is typically annotated by experts in that domain to ensure accuracy.

Within the medical domain [12], NER and RE are determinants for extracting relevant information from sources like EHRs, scientific literature, and other forms of medical data. Developing a tool that is both reliable and accurate for these tasks can considerably enhance healthcare service efficiency and enable more informed decision-making by healthcare providers.

This work is framed within the context of NLP aimed at languages with limited resources such as Spanish, Italian or Basque, particularly focusing on Information Extraction through NER and RE. Given the lack of resources for such languages, we aim to create robust models capable of effectively processing and understanding these languages, despite their limited resources. The ultimate objective is to advance information extraction capabilities in such environments.

1.2. Objectives

The primary objective of this project is to develop a solution based to adapt an existing NER model and a RE model for documents in Spanish with additional experimentation with documents in Italian and Basque.

Both models have been trained on EHRs with the objective of extracting laboratory examinations (referred to as EVENT) and measurement outcomes (RML - Results Measurement and Labs), and the relation between them. The main challenges presented by these tasks include the high complexity and variability of medical language, the lack of precision when identifying entities and relationships, and the requirement to handle amounts of data effectively.

We primarily focused on developing two different pipelines for each of the tasks by EHRs. This involves a dedicated pipeline for NER and another one for RE, aiming to simplify the production process. While these pipelines are trained independently, the RE pipeline is dependent on the output of the NER pipeline for annotating entities. Only after these entities have been annotated, the RE pipeline can effectively perform its function. These pipelines will be evaluated on their adaptability, specifically their ability to modify the base of the language model, as well as their effectiveness in extracting entities and relations.

The general workflow of this system would consist in taking the raw text of clinical cases as input and generating a list of pairs of entity mentions for which there is a relationship in the text. The output of the system includes both: the input document with entities well annotated and the annotated relationships in tabular form. The resulting tool should enhance the efficiency of healthcare services, aid in clinical decision-making processes, and potentially contribute to advances in medical research, especially in Spanish-speaking areas.

1.3. Document Structure

The rest of the document is structured as follows: A review of the state of the art for the entity and relation extraction task on the medical domain is included in chapter 2. Then, chapter 3 describes how the experimental part has been conducted in terms of datasets used, preprocessing, model architectures, and experiments. Chapter 4 summarises the evaluation metrics, the results obtained, and provides a brief discussion about problems encountered. Finally, in chapter 5 we share lessons learned and discuss future work.

Capítulo 2

State of Art

The rising volume of digital data being generated across sectors like biomedicine, finance, or social media, has increased the relevance of Named Entity Recognition and Relation Extraction tasks. Transformer-based models, outperforming traditional methods, have set new benchmarks in these fields. Our review aims to explore the latest developments in these transformer-based techniques and assess various advanced models specifically trained for information extraction tasks in the medical and biomedical domains.

In this chapter we will review transformer-based techniques that have had an impact on the two tasks analyzed in this work (NER and RE), paying special attention to the application of these tasks in the medical and biomedical domain.

2.1. Language Models

One of the technical advancements that have significantly influenced NLP, and by extension, NER to make IE, is the introduction of attention mechanisms.

Based on [13] and [14], among others, emerges Transformer-based models [4] use a self-attention mechanism to assess the relevance of each input token for every output token, capturing dependencies independent of their sequence positions. Each token in self-attention attends to all others, calculating a weight that indicates their importance, creating a new token representation as a weighted combination of all tokens. This enables the model to capture diverse linguistic structures.

Transformers are distinctive due to their use of multi-head attention - multiple self-attention layers stacked together - which allows the model to recognize various types of data relationships and dependencies in the data. They also use a Feed Forward Network (FFN) to sequentially process each output codification with prior ones.

Masked self-attention is used in the decoder to maintain the autoregressive property of the task, ensuring that each output token prediction can only depend on earlier output sequence positions. Such models apply varying 'attention' levels to different words during text processing.

For our specific NER and RE tasks in Spanish, Italian, and Basque, we chose distinct transformer models as our foundational models.

2.2. Transfer learning

Once we know a little about the mechanisms of attention and how transformers work, we can go deeper into models that leverage **transfer learning**. Transformer-based models are pre-trained on a vast corpus of text and subsequently fine-tuned on a specific task, like NER or RE. This strategy reduces the need for having extensively annotated datasets, speeds up the training process, and often solutions are in enhanced performance, particularly in domains with limited data.

In this way, the study by [15] examines NER methods, with SpaCy transformers achieving average results among other tools. [16] demonstrated superior performance of fine-tuned BERT models for relation extraction to handle clinical texts.

2.3. Multilingual language models

Multilingual models, as the name suggests, possess the ability to process and understand multiple languages—a critical attribute in the globalized digital era where data is derived from a wide range of languages and dialects.

Transformer-based models are usually trained on monolingual corpora, but several have been trained on multilingual corpora. Using multilingual models like Multilingual BERT (mBERT) or XLM-RoBERTa, we can prevent the loss of valuable insights due to language barriers.

These models are particularly beneficial in areas such as social media, where content is generated in a multitude of languages, especially in domains with limited data.

2.4. Named Entity Recognition (NER)

In the realm of the biomedical domain, BERT model has been trained to extract entities (NER) within a machine-reading comprehension framework [17]. The work, spanning six different datasets, achieved state-of-the-art performance across all. The study also compared different models such as other BioBERT variations, with the proposed method, outperforming the others. This framework consists on involves training a model to understand, interpret, and answer questions about a given piece of text [18].

[19] suggested a deep learning model for the development of an end-to-end system for extracting family history (FH) of patients information from clinical narratives, a crucial task given the importance of FH in risk assessment for numerous diseases, with a comparative analysis carried out between Long Short-Term Memory (LSTM), BERT, and ensemble models through majority voting.

Continuing with NER in the biomedical domain, we find works such as the ones presented in [20] and [21]. These explore methodologies for extracting medical and clinical entities from Chinese electronic health records (EHRs) to construct knowledge graphs and identify clinical entities, respectively. They address the challenge of irregular punctuation use in these records, which affects classification accuracy, suggesting a solution in the form of a BERT-based punctuation correction calculation. Both studies employ machine learning models, with [21] specifically being superior

in comparison with a bidirectional LSTM-CRF deep learning model over a traditional Conditional Random Field (CRF) model for recognizing clinical entities.

2.5. Relation Extraction (RE)

Another recent study introduces PARE[22], a novel, efficient method for Supervised Distance Relationship Extraction (DS-RE). PARE's process involves individually encoding sentences from entity pairs and unifying them into a single passage for BERT or mBERT (for multi-language settings) encoding. The study evaluated PARE on four datasets in both single-language and multi-language contexts, yielding consistent results applicable across various environments. By generating a unified, relation-aware summary using an attention mechanism, PARE effectively predicts the validity of relationships between entities.

The majority of RE tasks focus on identifying connections between entities within the same sentence, though there are instances of RE tasks operating across multiple sentences. For each instance, the entire text, whether it be the whole sentence or paragraph, is utilized as input to the model. [23] set a new benchmark for temporal RE in the clinical domain using both pretrained domain-specific models and a fine-tuned version of BERT. This highlights the efficacy of these methods in tackling temporal RE within the clinical context.

In [24], the authors evaluated the performance of BERT against a random forest, utilizing a variety of features. The authors employed BERT-based models like BioBERT and Clinical BERT for the task of drug-disease relation extraction from biomedical and clinical texts, thereby comparing the effectiveness of these models with the traditional Random Forest algorithm.

Finally, we will refer to the work by Soares et al. [25]. The authors present a training method, 'matching the blanks', aimed at generating meaningful relation representations from text using only entity resolution annotations. Paired with a new architecture for fine-tuning relation representations in BERT, the model delivers state-of-the-art performance in three relation extraction tasks, surpassing human accuracy in few-shot relation matching, and proves particularly effective in low-resource environments.

Capítulo 3

Materials and Methods

In this section, we delve into the crucial components and processes that were used in this study. We start with a review of the indispensable libraries and tools that facilitated our investigation research. We will then shed light on the datasets that served as the base of our research, primarily focusing on the E3C corpus. From this dataset, EHRs are obtained, which are called TESTLINK and CLINKART. EHRs in three languages are obtained from the evaluation task as mentioned in Chapter 1.

An indispensable phase of our study involved the preprocessing of these datasets, a process that ensured the data was properly adapted and structured for the subsequent stages. The adaptation of models, a vital segment of this research, is being made thanks to the integration of two major libraries: SpaCy and Hugging Face.

SpaCy is an open-source library for advanced NLP tasks in Python, designed for large-scale information extraction tasks with deep learning capabilities. Hugging Face is a leading company offering an open-source library that provides an open-source library for state-of-the-art NLP models, pre-trained on a large corpus of text data. These libraries provided us with powerful, foundational models that greatly contributed to the outcomes of our study.

Finally, we detail the experimental procedures that we employed. These experiments were specifically designed to unravel the intricacies of language model performance across different languages, including Spanish, Basque, and Italian, as well as combinations of these languages. The findings from these experiments, including testing French models with the Basque dataset, represent the core of our discoveries in this study.

3.1. Technical Setup

This research was conducted using Google Colaboratory, a popular cloud-based platform that provides an environment for machine learning and data analysis. This tool, often abbreviated as Colab, offers an interface for executing Python code.

The primary benefit of Colab, and the reason for its selection for this research, is its seamless access to powerful hardware accelerators such as Graphics Processing Units (GPUs). The work conducted during this study was specifically powered by a

GPU model T4, a high-performance, energy-efficient GPU by NVIDIA with a large RAM memory capacity.

For efficient version control and collaboration, the code and related resources developed during this research have been hosted in a dedicated GitHub repository. GitHub, an industry-standard platform for code hosting and collaborative software development, offered the team an easy-to-use interface for code sharing, issue tracking, and version control. This repository provides full transparency of the conducted work, and also serves as a comprehensive resource for other researchers who might want to replicate our experiments or build upon our findings. This practice tries to align with the principles of open science such as FAIR, promoting the accessibility and reproducibility of our work. The GitHub repository can be accessed publicly via the following URL: <https://github.com/JPardo08/PAQUITA.git>.

3.2. Libraries

The integration of SpaCy models with Transformers, specifically through the Hugging Face library, represents a significant development in the field of NLP. SpaCy supports over 72 languages and 80 trained pipelines for almost 24 languages [6].

SpaCy is an advanced open-source library for NLP known for its efficiency and ease of use. Its pipeline architecture allows in once a flexible integration of various NLP tasks including tokenization, part-of-speech tagging, dependency parsing, named entity recognition, and relation extraction. By integrating Hugging Face's trained transformer models into the SpaCy pipeline, it's possible to combine the contextual understanding of Transformers with the linguistic knowledge encapsulated in SpaCy's NER or RE layers, forming a comprehensive NLP solution.

Furthermore, SpaCy operates with its internal library known as Thinc, a unique machine learning framework that combines the adaptability of PyTorch with the high-performance capabilities of TensorFlow. TensorFlow and PyTorch are two popular open-source machine learning libraries that provide tools and frameworks for creating and training a wide variety of neural network models. This gives SpaCy an edge in deploying flexible and efficient machine learning models. Hugging Face, in turn, works directly with both TensorFlow and PyTorch, among others, further enhancing the robustness and versatility of the models it supports.

Prodigy is an annotation tool that integrates seamlessly with spaCy but can also be used with other deep learning tools and libraries. Prodigy aims to address the lack of a clear and repeatable methodology for annotation in the machine learning ecosystem, thereby streamlining the procedure. Data is adapted to be similar to the Prodigy's output.

Hugging Face, on the other hand, as a pioneering IT company, specializes in developing and providing transformer-based environments, datasets, and models dedicated to NLP and other AI fields. Transformers have greatly improved the performance of various NLP tasks such as NER, sentiment analysis, text classification, and more. These models capture the context of every word in a text, independently of its position, making them highly effective for understanding natural language.

The alliance of SpaCy's Thinc, TensorFlow, PyTorch, and Hugging Face, 3.1 forms a good framework, empowering the creation of advanced machine learning models.

This synergy allows users to leverage the best aspects of each technology.

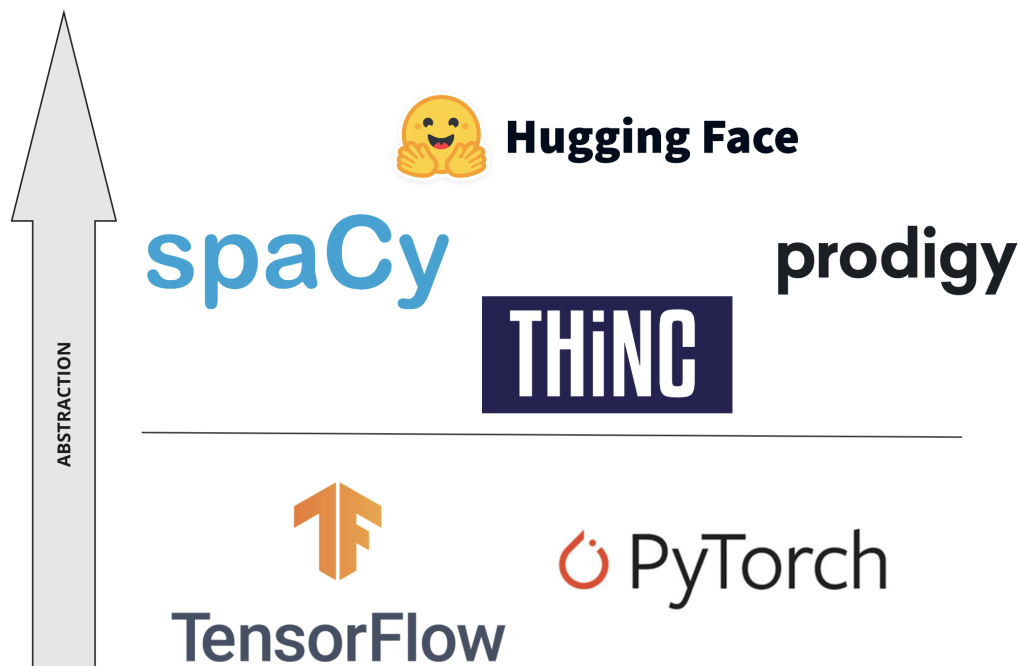


Figura 3.1: Libraries.

3.3. Datasets

A critical aspect of building effective RE models is the availability of high-quality datasets that enable model training and evaluation.

A high-quality dataset for NER and RE is characterized by accurate annotations, robust and diverse examples, and a representation of complex relationships. It should be extensive enough to include several scenarios and contexts, covering a comprehensive range of entities and their possible relations.

It should also be annotated by domain experts to ensure the precision and relevance of the information. In the medical domain, it often includes annotations of medical terms, their relationships, and contexts from sources like EHRs and other medical databases.

An understanding of the datasets used in our study is essential for interpreting the results and conclusions of our experiments accurately.

Moreover, examining each dataset helps identify potential biases, anomalies, or gaps in the data, enabling us to validate the robustness of our models and improve their performance. It also helps ensure that our methods are applicable and effective for the diverse scenarios and contexts represented in the datasets as well as the limits they could carry.

A clear example would be a research context where we are working with two datasets: one contains Electronic Health Records from a hospital in Spain, and the other contains medical research papers from an Italian university.

In the Spanish EHR dataset, the data might be biased towards certain diseases or conditions that are present in that specific hospital’s patient population. This could lead to biases in our models, making them better at recognizing entities and relations associated with those conditions but worse at handling others. By carefully examining this dataset, we could identify such biases and take measures to mitigate them, such as adding to the dataset additional data in order to enhance either entities and relations.

Similarly, in the Italian research paper dataset, there might be gaps in the data. For instance, it might lack information on certain types of relationships between entities that are not often discussed in academic literature. By identifying these gaps, we could improve our models by ensuring they are trained on a more comprehensive and representative set of relations.

Furthermore, each dataset could contain anomalies, such as unusual uses of language or rare entities and relations, that could affect the performance of the models. By identifying and understanding these anomalies, we could better fine-tune our models to handle them.

Finally, by understanding the diverse scenarios and contexts represented in each dataset - clinical practice in the Spanish dataset and academic research in the Italian dataset - we can ensure that our models and methods are not only applicable and effective in these specific scenarios but also have the ability to generalize to other, similar scenarios. This also gives us an idea of the limits and boundaries within which our models are expected to perform efficiently.

In this section, we will provide an overview of the datasets used in our study. We will describe the characteristics and properties of each dataset.

3.3.1. E3C corpus

The **E3C corpus** [26] is a collection of semantically annotated clinical texts that are available for free and can be used to analyze, benchmark, and train information extraction systems. It is a multilingual corpus that includes text in English, French, Italian, Spanish, and Basque. The corpus contains two types of annotations: clinical entities, such as pathologies, and temporal information and factuality, including events.

The dataset is structured into three folders:

- **data_annotation**: contains the annotated relationships in '.xml' files, where the 'id' field indicates the origin and the 'target' field indicates the destination of the relationship. These annotations are divided into two layers, namely Layer 1 and Layer 2.
- **data_collection**: contains the text documents in the form of '.json' files, some of which are extracted from SPACCC. The documents are divided into three layers corresponding to layers 1, 2 and 3 of the dataset. Layers 1 and 2 match with the data_annotation folder.
- **data_validation**: contains the annotations for the validation documents in the dataset, only for Layer 2. However, not all layers have annotations for validation, only those that are referenced.

- **documentation:** includes two items: a guide that explains how the documents have been annotated, and a '.txt' file containing the documents for training. If a document is not included in the training file, it will be used for testing. It seems that layer 3 is reserved for testing purposes.

3.3.2. TESTLINK & CLINKART CHALLENGE

Based on E3C corpus 3.3.1, we have TESTLINK and CLINKART databases¹ as it is shown in 3.2. By them, the aim is to identify test and measurement results mentioned in the text. Then, link them to the corresponding laboratory tests (EVENTS) and measurements (RML). This requires first the identification of both elements in the text and the establishment of a relationship between them. The TESTLINK database comprises of 81 documents in Spanish and 90 in Basque, while the CLINKART database is home to 83 Italian documents.

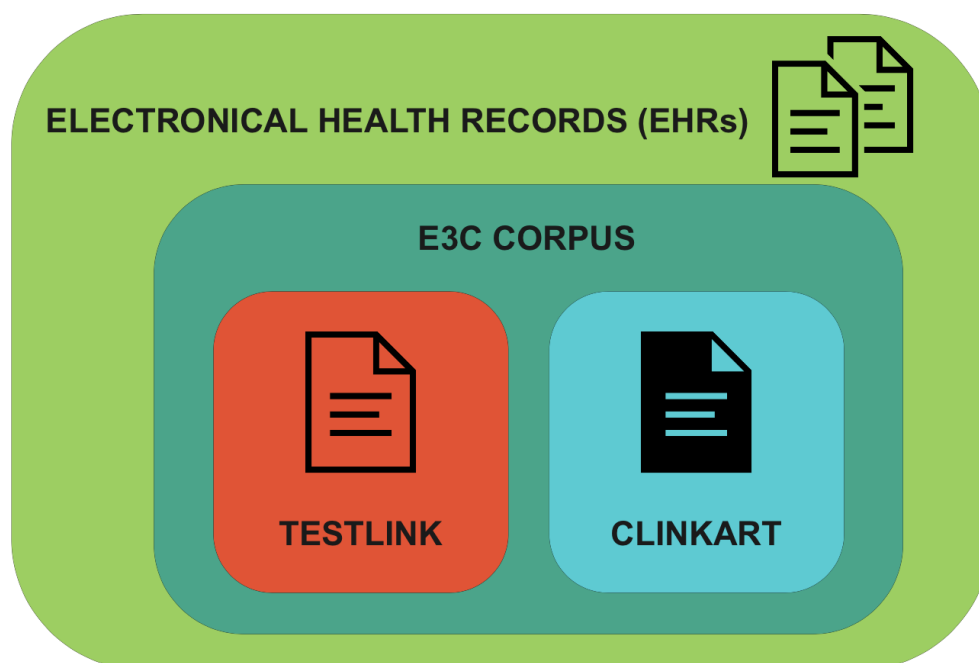


Figura 3.2: Electronic Health Records (EHRs)

All relationships in this context are of the PERTAINS TO type. These relationships can be one-to-one, due to the nature of the data.

In the dataset, each document is placed on a separate line, and a blank line is used to separate one document from the next:

- DOCID: document identifier.
- t: marker to identify the lines that contain the text of the documents.
- TEXT: document text.

¹<https://e3c.fbk.eu/testlinkiberlef> and <https://e3c.fbk.eu/clinkart>

Every annotated relationship is represented as an ordered pair of entity mentions (i.e., RML, event) and is placed on a distinct line. The entity mentioned in the relationship are identified by their initial and final character offsets, and the length of the mentioned text may be specified but is not obligatory:

- DOCID: document identifier
- REL: marker to identify the lines containing the relations of the given document.
- RML_START: displacement of the initial character of the RML entity mention in the document.
- RML_END: the final character of the RML entity mention in the document.
- EVENT_START: the initial character of entity mention EVENT in a document.
- EVENT_END: the final character of the mention of the entity EVENT in the document.
- RML_TEXT [optional]: text space of the entity mention RML.
- EVENT_TEXT [optional]: extension of the text of the entity mention EVENT.

On the other hand, we have a folder which contains one '.tsv' file for each DOCID contained in that dataset. This file, in order to make the task easier, has the text tokenized.

- Phrase: id of the phrase inside of the paragraph.
- Id: id of the word inside of the phrase.
- Start: token start inside of a word.
- End: token end inside of a word.
- Word: word. In this case it would be a simple word, a punctuation sign, or a measure.

3.4. Preprocessing task

The process of cleaning and structuring data from unstructured to structured format for NER is quite different for RE. Both of them involve several steps, divided into two main blocks: data loading and cleaning data.

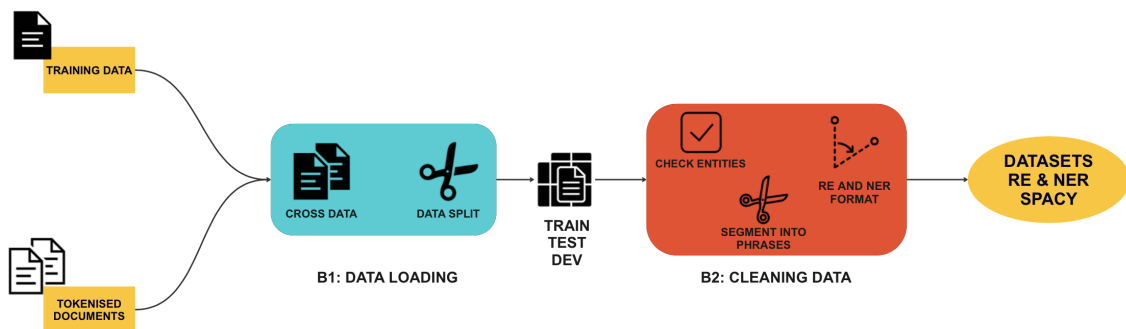


Figura 3.3: Preprocessing workflow.

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Figure 3.3 demonstrates the preprocessing stage of our methodology, which is structured into two principal components: data loading and cleaning data.

We start with the first block, Data loading. Here, we collect unstructured medical text data from the sources specified before such as Testlink and Clinkart databases.

Time to load the training data. It is separated into two sets: one containing the table format with each entity and relationship annotations, the other with the documents. The data were tokenized by default, so we directly load the tokenised data.

Tokenization is the process of breaking down the text into individual words or tokens. This is a crucial step as each token will be analyzed and classified individually in NER.

The annotation data are integrated into the documents registry by aligning the annotations with their corresponding phrases, based on the DOCID indicator. This reveals that some documents lack annotated relationships, so these documents are discarded.

The EVENT and RML columns are restructured, transforming them into four distinct columns, two representing each entity type. Within these, one column signifies the starting point of the span, while the other denotes its end.

The dataset is divided into subsets for training, testing, and validation, determined by the DOCID.

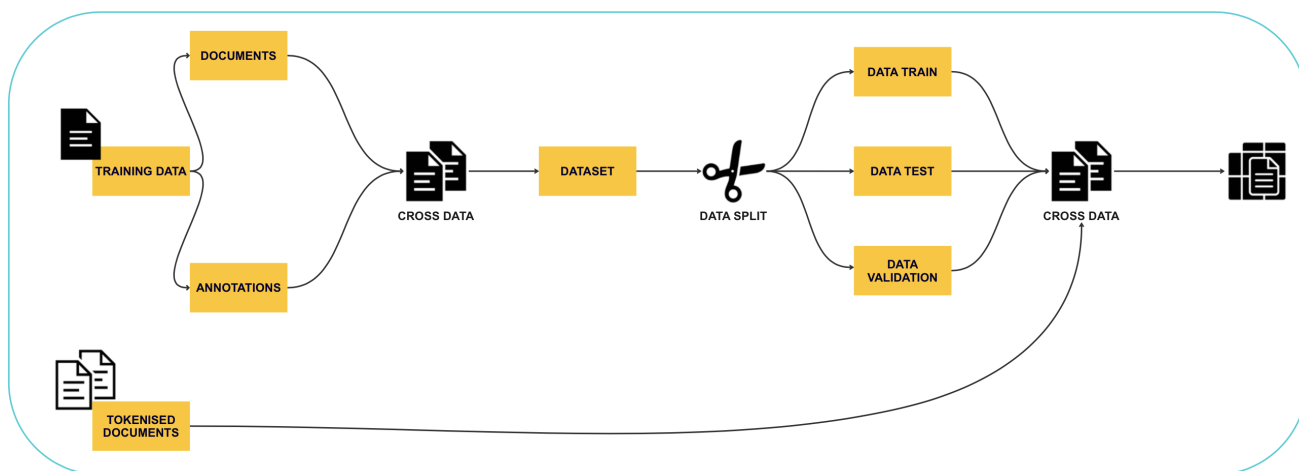


Figura 3.4: First stage of the preprocessing tasks: Data loading.

Figure 3.4 represents the initial stage of the preprocessing procedure. The datasets, represented in yellow, are the outcomes of specific operations.

Now, the second block, Cleaning data. By examining the tokenisations chosen for each dataset, we determine which words correlate to an entity.

The dataset is dissected by DOCID, representing different levels according to documents, followed by phrases and words to identify entities within each phrase of each document. We realize that not every sentence within a document carries annotations, so these sentences are discarded.

Then, each one of the split sets is integrated with tokenised data to match the previous records based on the DOCID.

The data is sorted based on the phrases. In all sentences containing entities, there seems to be at least one entity of the EVENT type, followed by one or more of the RML type. This observation suggests that the relationships typically exist within the same sentence rather than dispersed throughout the text.

Data is transformed from documents into shorter phrases. This collection of sentences is then processed to generate the final json, carrying the relevant data which serves to get the final binary SpaCy file.

The RE json has as keys: 'document', representing each sentence; 'relations', representing each relationship within a sentence, detailing the child, the head at the word level, and the type of relation; and 'tokens', illustrating each entity within the sentence, indicating the span at the token and word level that is the actual text word and the entity label within the sentence.

Following this, the NER data undergoes conversion to BILUO format from IOB via a command-line interface (CLI) command, with the aim to generate the NER json analogous to that of the relationships. Once this json file is acquired, it is converted into a binary SpaCy format.

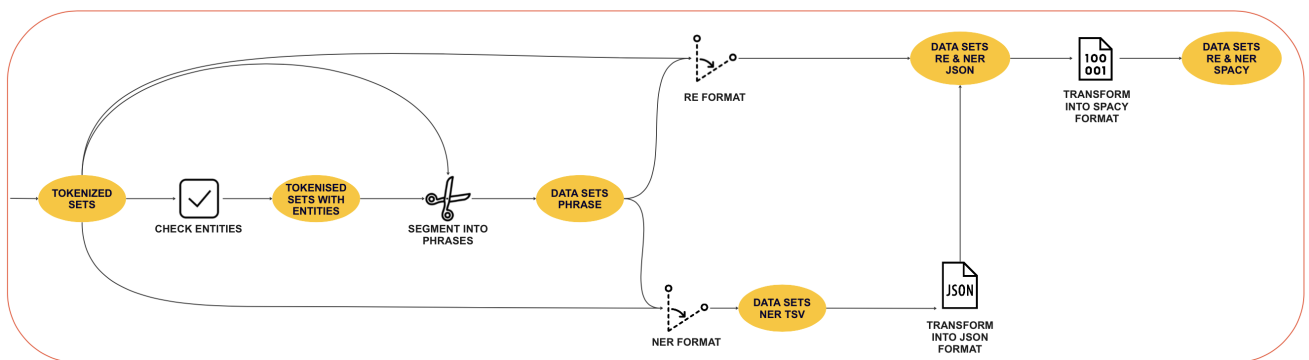


Figura 3.5: Second stage of the preprocessing tasks: Cleaning data.

Figure 3.5 showcases the second phase of the preprocessing steps. The data, symbolized as a circle, includes three sets: train, test, and validation. Various actions are applied on these sets.

The annotation process for NER is known as IOB (Inside, Outside, Beginning) format. This is a common format used in NER to label tokens by their entity type and their position inside the entity.

'B-' prefix is used before a tag if the token is at the beginning of an entity, 'I-' if it is inside but not at the beginning, and 'O' if the token does not belong to any entity. For example, in the phrase '1.4 mg/dl', '1.4' would be labelled as 'B-RML' and 'mg', '/', and 'dl' as 'I-RML'. This step can be intensive and may require medical expertise to ensure accurate annotations.

BILUO is a common tagging scheme used in NER, and it stands for Beginning, Inside, Last, Unit, and Outside. It is an extension of the IOB (Inside, Outside, Beginning) format, which is also used in NER. BILUO, however, provides more granular detail, which can be particularly beneficial for multi-token entities. In this case, for example, 'dl' would be 'L-RML' and the EVENT, as it is a single word, would be 'U-EVENT'.

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The annotation for RE is different. After identifying entities, the next step was to identify the relationships between these entities. For this, a separate label is created for each type of relationship. In this case, we only have one, concretely REL which means PERTAINS TO relationship.

For example, a relationship between a RML and an EVENT is labelled as 'REL'. The relationships are typically identified and labelled at the sentence or phrase level. To visualize it, in the sentence 'Paciente de 65 a. de edad, que presentaba una elevacion progresiva de las cifras de PSA desde 6 ng/ml a 12 ng/ml en el ultimo ano.', the relationship between '6 ng/ml' or '12 ng/ml' as RML and 'PSA' as EVENT would be labeled as 'REL'.

Once these steps are complete, you will have a structured dataset that can be used to train NER and RE models. The models can then be used to automatically extract entities and relationships from new, unstructured medical text data.

Experiments	Train Sentences	Test Sentences	Dev Sentences
Spanish	118	98	46
Italian	156	71	53
Basque	261	144	74
Spanish-Italian	274	169	99
Spanish-Basque	379	242	120
Basque-Italian	417	215	127
All	535	313	173

Cuadro 3.1: Number of sentences in NER experiments.

From the table 3.1, key points are:

- The Basque language experiment has the highest number of training, testing, and development sentences among the single language experiments. This implies that the Basque language experiment was conducted with more data compared to the Spanish and Italian experiments.
- The All experiment, which presumably combines all the languages, has the highest number of sentences in each category. This suggests that a more diverse data set was used in this case.
- The Basque-Italian experiment has the second-highest number of sentences in each category, indicating a substantial amount of data was available for this bilingual experiment. This could potentially lead to better model performance due to the availability of more training data.
- The Spanish experiment had the least number of training, testing, and development sentences among all experiments. This might limit the performance of the model for the Spanish language, due to less data for training.

Based on the data presented in the table 3.2, we can conclude the following:

- The experiment with the most train sentences is Basque-Italian, followed closely by Spanish-Basque. The experiment with the least train sentences is Spanish. This indicates that the models trained on Basque-Italian, or Spanish-Basque data had the most information to learn from.

Experiments	Train Sentences	Test Sentences	Dev Sentences
Spanish	262	98	46
Italian	280	71	53
Basque	476	143	73
Spanish-Italian	542	169	99
Spanish-Basque	738	241	119
Basque-Italian	756	214	126
All	615	242	172

Cuadro 3.2: Number of sentences in RE experiments.

- Conversely, the Spanish experiment had the fewest test sentences, whereas the All experiment used the most. This suggests that the models were tested more rigorously when data from all languages were combined.
- Regarding development sentences, again the Basque-Italian experiment is the highest value, while Spanish had the fewest. This might influence the final tuning of the models, as more development data generally allows for more reliable validation of the model's performance during the tuning process.
- In the All experiment, the amount of data used for training, testing, and development is not the highest compared to other experiments. This could imply that in multilingual experiments, a balanced dataset coming from different sources might be more important than the volume.

3.5. Models

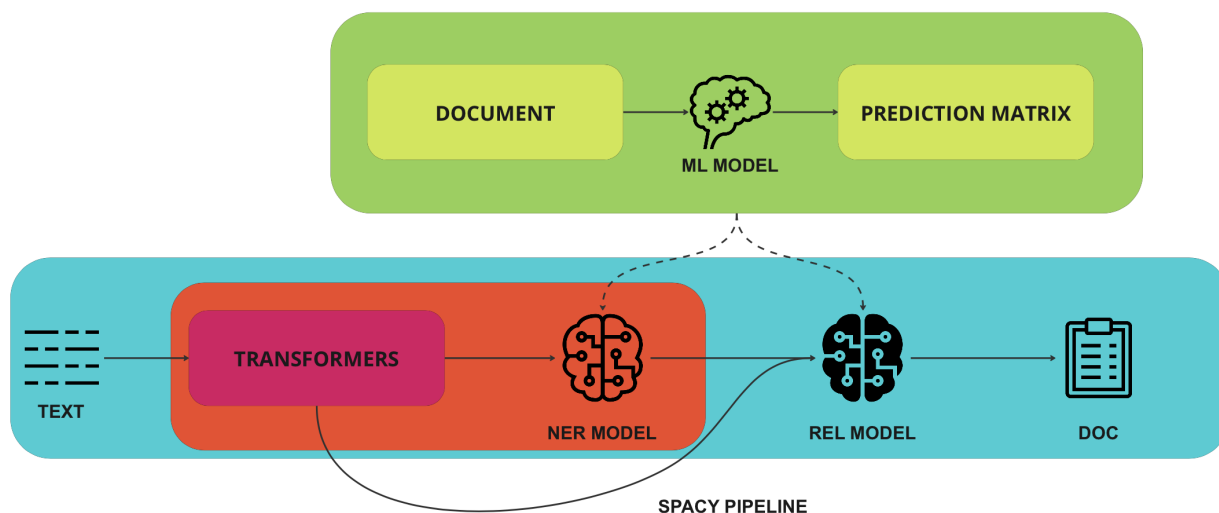


Figura 3.6: NER and RE SpaCy Pipeline.

The core of this work is to use transformed-based models for embedding. These models are known for their ability to use the context of text using something called self-attention mechanisms, explained above.

As 3.6 shows, each constructed model has two main layers: the transformer and the

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NER or RE layer. The transformer layer tokenizes input text before passing it to the NER or RE layer. Consequently, the pipeline could be either a transformer and NER component or a transformer and rel_extractor component.

In the relation extraction model is 'TransformerListener' that connects to the transformer component to get token vectors. This allows multiple components to listen to the same transformer model and pass gradients back to it, enabling multi-task learning. As Transformer models provide wordpiece vectors, a pooling layer 'reduce_mean' is needed to get token vectors from word pieces.

The config file, is where we define the language and pipeline components, outlines intricate object trees which are constructed and validated incrementally during training.

The spaCy registry is used to register all the important functions. This registry allows the functions to be referenced in a configuration file, which makes the model easy to train and customize.

In none of the cases, NER and RE are not together as a unique model. Instead, they are separated models. To use them, we have to call individually.

Speaking in terms of implementation in spaCy, the developers aim to keep the application programming interface (API) and usage as simple as possible.

Users can iterate over the entities in a document to obtain a Span object, providing the start and end positions of each entity and a way to iterate over the tokens. Once the entities are known, is time to classify is there is any relation between them.

3.5.1. NER

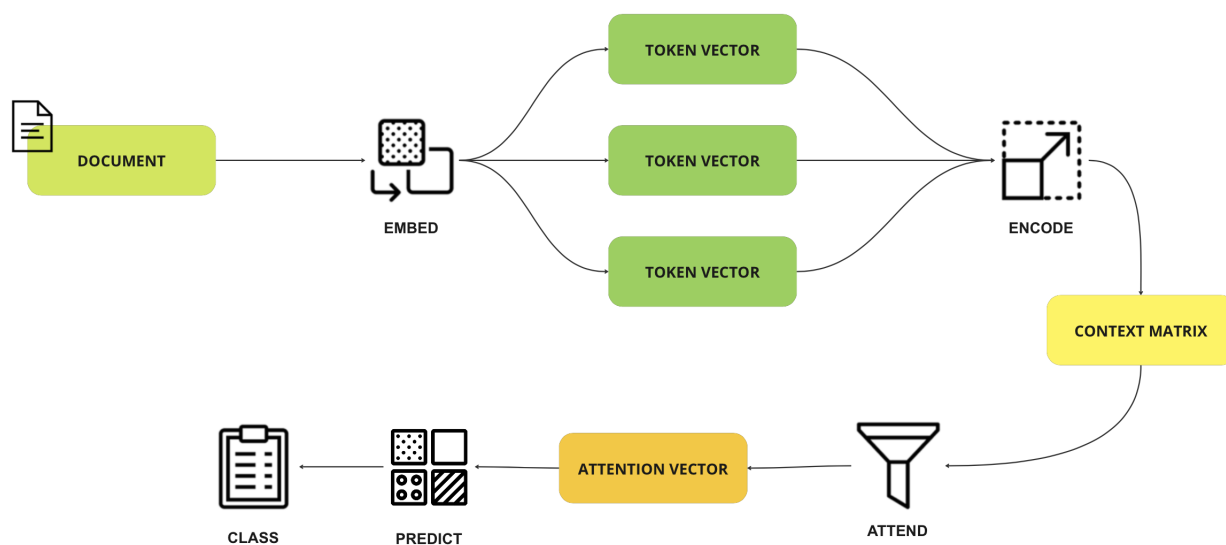


Figura 3.7: SpaCy NER process.

The adopted model follows a transition-based structured prediction approach, where the sentence is processed as a state machine. This approach involves maintaining a state and manipulating the structure by selecting from a set of available actions.

It has a statistical model used to predict transitions. This model, as 3.7 shows, has four actions:

1. Embed: represents words as dense vectors or embeddings.
2. Encode: recalculate embeddings based on their context.
3. Attend: representation is condensed into a summary vector or a single piece.
4. Prediction: make a prediction from the summary vector.

At the initial stage, the system uses techniques known as the 'hashing trick' [27] or 'bloom embeddings' [28] to generate word representations. In simple terms, the 'hashing trick' helps to transform each word into a more compact and computationally manageable form, while in 'bloom embeddings' a word is transmitted with multiple features, like its root form and part of speech, all combined into a single, rich representation. Both techniques help to process text data more efficiently.

In encoding phase, each outputting a vector, are combined to create another function that outputs a concatenated vector which pass through a multi-layer perceptron with a Maxout unit for the hidden layer.

To the attend phase, we embed individual words by Convolutional Neural Networks (CNNs) for dimensionality reduction - converting a matrix format into a vector format using multiple filters and striding operations. Residual connections are used to maintain similarity between the input and output vector spaces. Attention is used as a mechanism to create a summary vector.

To predict, a tensor of the embedded words is created. This tensor is fed into a tri-gram Convolutional Neural Network (CNN) resulting in context-aware word vectors. 'attention weights' are pre-computed and multiplied by the first hidden layer. A loop is initiated that continues until the end of the buffer is reached. This loop itself consists of computing the feature IDs for the current state, using a basic multi-layer perceptron to obtain action probabilities.

3.5.2. RE

Once the NER model is implemented, we explore the implementation of a custom component tasked with predicting the relationships between named entities.

Figure 3.8 provides an illustration of the structure of the RE process, outlining the steps it follows:

1. Each token in the sentence is translated into a vector. This can be done using a HuggingFace Transformer.
2. Candidate instances in the sentence are determined. Given the interest in direct, binary relations between two entities, there can be two relevant instances: one with the first RML, the second as the EVENT, and one with the reverse.
3. For each instance, the vectors of the two corresponding entities are concatenated. These simple vectors contain data representing the subject and the object.
4. All instance vectors are concatenated into one larger tensor, which becomes the input of the classification layer.

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5. The final output is designed to predict one relation type: REL, explained above. For each instance and for each relation type, a prediction is made. Values between 0 and 1 are given, with a threshold above 0.5 considered a 'True' relation.

The first layer of the model, which generates instance data from raw text, has a forward function and backpropagation. A 'reduce_mean' layer is applied to summarize multiple token vectors into one entity vector. 'get_instances' function defines the candidate instances in a document. It is used to pair up any two entities in a document within a certain maximum distance of each other or co-occur in the same sentence.

The classification layer takes an output dimension 'nO' and an input dimension 'nI' to return a model that converts a 2D matrix into another 2D matrix. For simplicity, this layer is defined as a linear layer followed by a logistic output activation to ensure predictions are within the [0,1] interval.

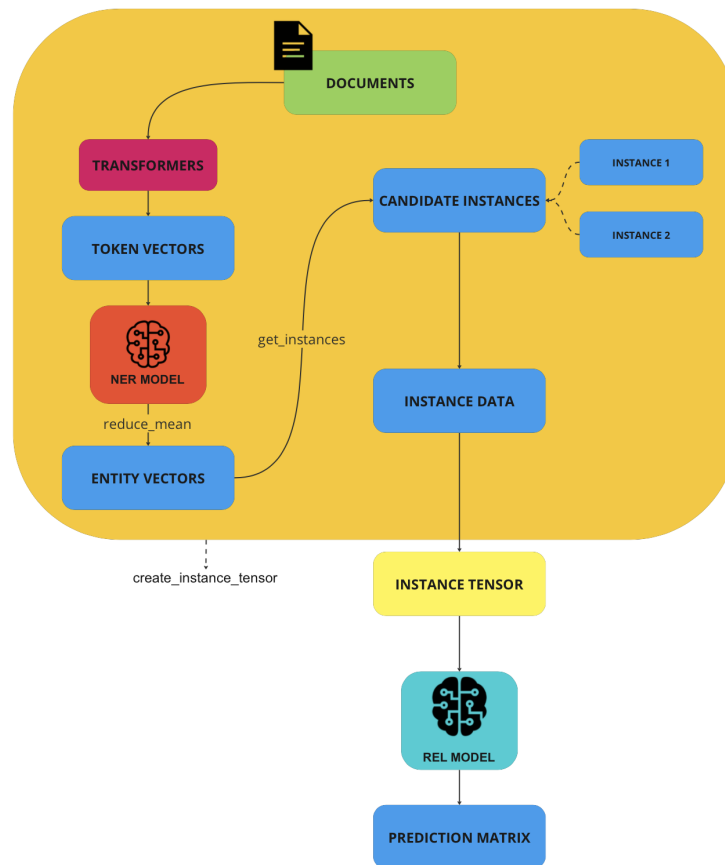


Figura 3.8: SpaCy RE process.

Consider the phrase 'Paciente presentaba biopsia negativa' as an example, as it is illustrated in 3.9. Initially, this phrase is processed by the transformer model for embedding, and subsequently, the NER model identifies 'biopsia' as EVENT and 'negativa' as RML.

Post entity recognition, we have token vectors, each row corresponding to a word with an n -dimensionality per row. For illustration purposes, we use $n = 5$. The vectors of the identified entities, EVENT and RML, are highlighted in green and orange respectively.

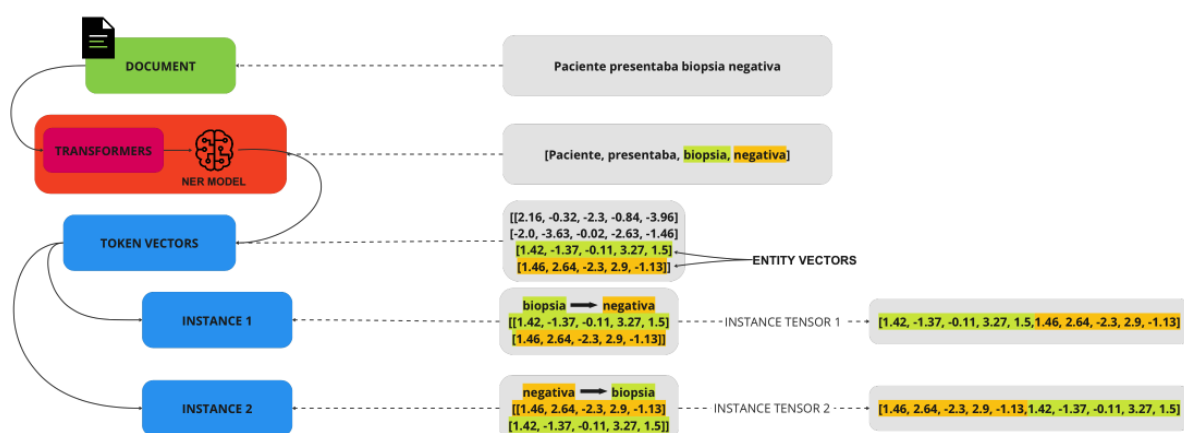


Figura 3.9: SpaCy RE process. Example 1.1.

Utilizing these token vectors, the 'get_instances' function is invoked to decipher the semantics of the phrase. Two interpretations are generated - one where EVENT is related to RML and the other where RML is related to EVENT. 'Biopsia' pertains to 'negativa', or 'negativa' pertains to 'biopsia'.

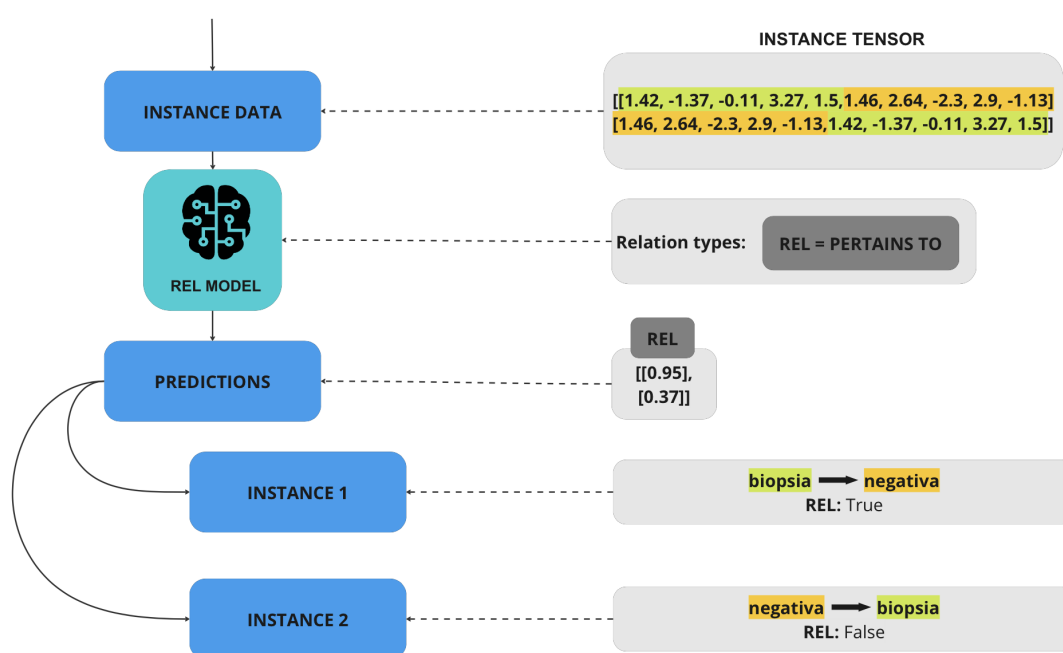


Figura 3.10: SpaCy RE process. Example 1.2.

For both semantic interpretations, an instance tensor is formed using the 'create_instance_tensor' function, 3.10. This tensor is then used as an input for the REL model's classification layer to assess if a REL relationship exists between the identified entities.

In the final prediction stage, a prediction matrix is generated. Each column of this matrix represents a class (in our case, there is only one - REL), while each row corresponds to an instance.

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In 3.11 is shown the sentence 'Paciente asciende PSA a 12 ng/ml' as an illustration. Initially, this sentence is sent through the transformer model for embedding, followed by the application of the NER model. This model recognizes 'PSA' as an EVENT and '12 ng/ml' ('12', 'ng', '/' and 'ml') as RML.

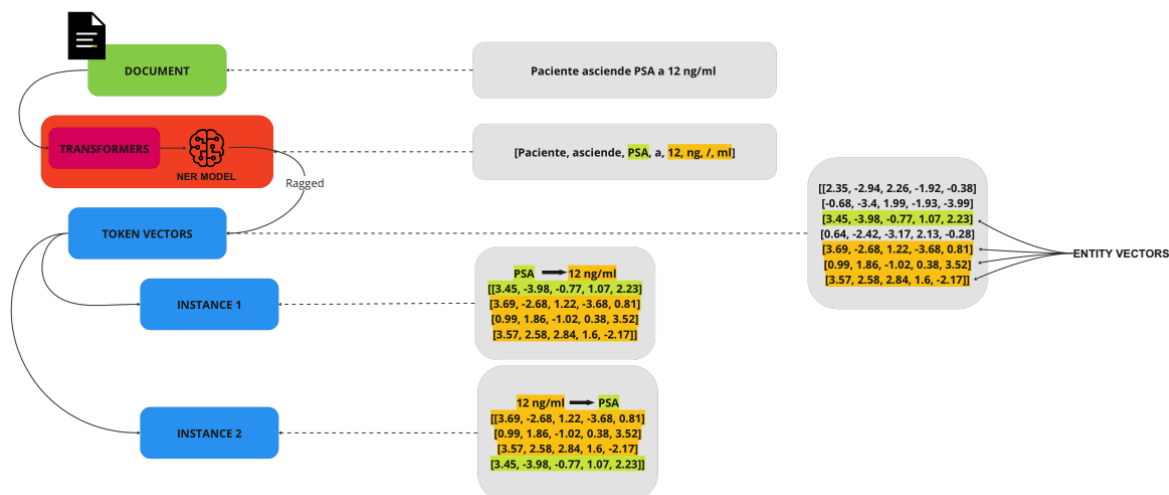


Figura 3.11: SpaCy RE process. Example 2.1.

Ragged, a unique data structure, is utilized at this point. It's particularly efficient when dealing with sequence data of variable length, managing to keep the data contiguous in memory for straightforward processing.

Following the entity recognition step, we obtain token vectors, each word corresponding to a row with an n -dimension per row, for this case, $n = 5$. The vectors for the recognized entities (EVENT and RML) are color-coded for distinction, with EVENT in green and each token of RML in orange.

These token vectors are then put through the 'get_instances' function to comprehend the sentence's semantics. It generates two possible understandings - one where EVENT relates to RML and another where RML connects to EVENT. 'PSA' pertains to '12 ng/ml' or '12 ng/ml' pertains to 'PSA'. To encapsulate multiple vectors into a single representative vector, the 'reduce_mean' function is applied.

These semantic possibilities are then converted into instance tensors using the 'create_instance_tensor' function. This tensor serves as input to the REL model's classification layer, determining whether a REL relationship between the identified entities exists.

Lastly, the prediction phase yields a prediction matrix, as 3.12 shows. This matrix's columns represent classes (in our scenario, there's just one - REL), and each row is associated with an instance.

Please note, these examples are provided purely for illustrative purposes. The numbers associated with token vectors and matrices do not accurately reflect real-world scenarios.

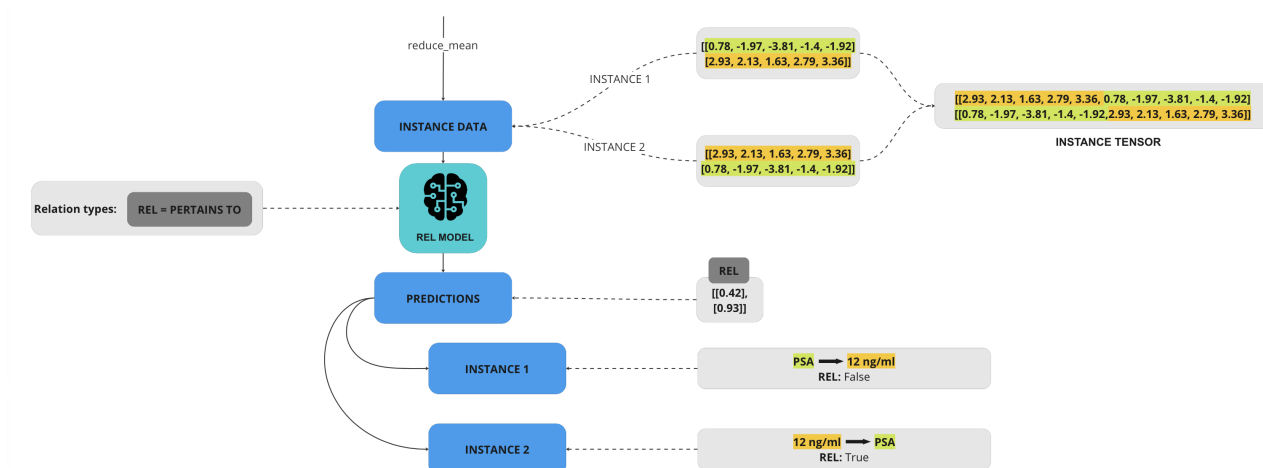


Figura 3.12: SpaCy RE process. Example 2.2.

3.6. Experiments

In this section, it is explained the experiments made to train NER and RE models. These attempts are crucial in our overall objective of extracting clinically relevant relationships from unstructured medical text data. How it is mentioned above, to see the applicability of the study, these experiments have been made over texts in three languages: Spanish, Italian and Basque.

The experiments are conducted by individually training the NER and RE models for each language, then by combining pairs of languages, and finally by mixing all the languages at the same time.

For the second and third modalities of experiments, the datasets created are shuffled. This implies that the order of the data entries is randomized, ensuring a diverse mix of phrases during the training process. Shuffling is a crucial step in data preprocessing that helps improve model performance and avoid potential bias. This way, the model gets a chance to learn from a varied representation of the data in each epoch.

Ensuring the accuracy of training data is crucial. If the selected span in the training data does not match the actual sentence, it could lead to errors, claiming that no instances could be identified in that sentence.

Another consideration to be noticed is the implications of English-centric models and their negative impact on other languages, particularly, in this case, with Spanish, Italian and Basque. It is crucial to be conscient of the potential bias created by the foundational models, like those provided by Hugging Face, which is primarily based on English for base models.

Managing Basque language can be challenging due to the absence of pre-trained models in SpaCy. Thus, I attempted to use a similar language who has certain linguistic similarity to Basque, French.

Each experiment in our study, indistinctly in NER o RE, primarily utilizes a Hugging Face transformer model known as 'xlm-roberta-base'². This model is a variant of the

²<https://huggingface.co/xlm-roberta-base>

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RoBERTa transformer, specifically designed for multilingual tasks. The 'xlm-roberta-base' model is trained on a wide variety of languages and can understand and process text data in multiple languages simultaneously, making it a robust choice for our diverse language experiments.

However, we take a different approach for the Spanish experiment. Here, we employ a model named 'PlanTL-GOB-ES/roberta-base-bne'³. This model is an adaptation of the RoBERTa model, but it is specially fine-tuned on the Spanish language. It was trained using the Biblioteca Nacional de España (BNE), which provides a rich source of diverse Spanish-language text data from 2009 to 2019. This specific focus on Spanish makes the 'PlanTL-GOB-ES/roberta-base-bne' model particularly effective for tasks involving Spanish language text data, hence its use in our Spanish experiment.

The learning rate for all experiments is initially configured to be 0.00 by default. As the experiments progress, this learning rate is updated accordingly. The 'xx' denotation in the SpaCy Language column indicates SpaCy base is multilingual in nature. The following table are presented a summary of the experiments 3.3:

Experiments	Transf. model	Spacy language	Datasets
Spanish	roberta-base-bne	es	TESTLINK
Italian	xlm-roberta-base	it	CLINKART
Basque	xlm-roberta-base	fr	TESTLINK
Spanish-Italian	xlm-roberta-base	xx	BOTH
Spanish-Basque	xlm-roberta-base	xx	TESTLINK
Basque-Italian	xlm-roberta-base	xx	BOTH
All	xlm-roberta-base	xx	BOTH

Cuadro 3.3: Experiments overview.

To visualize the final results of each experiment, I developed an interface using Streamlit. This interface allows for dynamic interaction with the data, offering users the ability to select specific languages for both tasks. Users can input any text and choose to visualize either the NER results, the RE results, or both.

The NER functionality highlights named entities over the user's text, assigning different colours to different types of entities. The RE functionality, on the other hand, identifies the relationships between these entities and presents the relationships and their corresponding confidence scores in a tabular format.

A sidebar contains controls for language selection and for choosing which results to visualize: NER, RE, or both. The interface is not only user-friendly but also allows for effective visualization and interpretation of results.

The Streamlit interface developed for visualizing the results of each experiment needs some enhancements, as it is reflected in 3.13. Although the interface is quite user-friendly and aids effective interpretation of results, it requires further refinement for better performance. Notably, the current threshold is fixed at 0.1 for visualizing all detected relationships, which might need adjustments for more precise results.

³<https://huggingface.co/PlanTL-GOB-ES/roberta-base-bne>

Paquita – A Named Entity Recognition and Relation Extraction Tool

Welcome to my NER and RE tool. Here you can input text and extract named entities as well as their relations.

Enter your text here

Paciente de 65 a. de edad, que presentaba una elevación progresiva de las cifras de PSA desde 6 ng/ml a 12 ng/ml en el último año. Dicho paciente había sido sometido un año antes a una biopsia transrectal de próstata ecodirigida por sextantes que fue negativa. El estudio espectroscópico de ésta lesión mostró una curva de colina discretamente más elevada que la curva de citrato, con un índice de Ch-Cr/Ci > 0,80, que sugería la presencia de lesión neoplásica, por lo que se biopsia dicha zona por ecografía transrectal. La AP de la biopsia confirmó la presencia de un ADK próstata Gleason 6.



Extract

Named Entity Recognition Results

Paciente de 65 a. de edad, que presentaba una elevación progresiva de las cifras de **PSA** EVENT desde **6 ng/ml** RML a **12 ng/ml** RML en el último año. Dicho paciente había sido sometido un año antes a una biopsia transrectal de próstata ecodirigida por sextantes que fue **negativa** RML. El **estudio** EVENT espectroscópico de ésta lesión mostró una curva de colina discretamente más elevada que la curva de citrato, con un **índice** EVENT de Ch-Cr/ **Ci** EVENT **> 0,80,** RML que sugería la presencia de lesión neoplásica, por lo que se biopsia dicha zona por ecografía transrectal. La AP de la biopsia confirmó la presencia de un **ADK** EVENT próstata **Gleason** EVENT **6.** RML

Relation Extraction Results

	Entity 1	Entity 2	Relationship
0	estudio	índice	0.3904532194137573
1	estudio	Ci	0.8297896981239319
2	estudio	> 0,80,	0.9941149353981018
3	índice	Ci	0.31696343421936035
4	índice	> 0,80,	0.9414492845535278
5	Ci	> 0,80,	0.4771077632904053

Figura 3.13: Interface visualization.

Capítulo 4

Evaluation

In this section, we will present the assessment measures employed for classification tasks in NER and RE. We will take a close look at the performance metrics used to quantify the effectiveness of our models, detailing how these metrics are calculated.

Further, we will explore the results obtained from each experiment conducted during this study, considering models trained on different languages and combinations thereof. We will discuss the outcomes, analyzing their implications and the insights gained from these explorations.

Lastly, we will discuss the challenges faced during the execution of these tasks. We will outline the limitations and difficulties encountered throughout the process, providing a comprehensive view of the practical issues that often arise in this project. This discussion aims to facilitate a more thorough understanding of the complexity involved in the tasks of NER and RE.

4.1. Evaluation Metrics

Models are evaluated using precision, recall, and F-score, as any other classification tasks. The model with the highest F-score is chosen for final evaluation on a reserved test dataset. Given the common data imbalance in relation extraction tasks, achieving high F-scores can be challenging due to the abundance of similar instances.

For an entity and relation prediction to be considered correct, both the entities involved in the relation must be correctly identified, and their order in the relation must be accurate.

- **Precision:** quantifies the fraction of accurate predictions from all predictions.
- **Recall:** calculates the fraction of accurate predictions among all actual relations.
- **F1 measure:** the harmonic mean of precision and recall and provides a balance between the two metrics.

4.2. Results

In this section, we present the outcomes of our extensive experimentation and analysis. Our primary focus has been to assess the efficacy of various transformer models

for NER and RE tasks, conducted across multiple languages. The results are a reflection of our experimentation process, where each experiment was designed with a specific transformer model and language in mind.

The insights drawn from these results will serve to illuminate the strengths and potential areas for improvement in our methodologies.

As we proceed through this section, we will first explore the detailed performance metrics from the training phase for both the NER and RE models. This will provide insights into the successes and challenges we encountered during the training process. Subsequently, we will present the evaluation results specifically for the relation extraction tasks, thereby offering a comprehensive understanding of the model's performance in the applied scenarios.

Upon completion of the training process, we obtain two distinct models. The first model represents the outcome of training continued until the last epoch. This model has been trained through the entirety of our specified iterations, learning from the full extent of our training data. The second model, on the other hand, is the one which demonstrated the best performance throughout the various epochs of training. This model is a reflection of a specific moment in the training process when our performance metric reached its optimum value, indicating the most effective learning state across all epochs.

These two models offer different perspectives on the performance and effectiveness of the training process.

Experiment	Steps	Loss Transf.	Loss NER	F-score	Recall	Precision
Spanish	1000	0.00	0.00	76.53	81.51	72.12
Italian	1200	42.13	52.17	77.52	78.12	76.92
Basque	800	65.04	125.54	89.58	89.83	89.33
Spanish-Italian	1000	68.54	116.37	81.09	81.00	81.19
Spanish-Basque	1000	72.05	150.18	87.55	89.40	85.77
Basque-Italian	1800	16.92	22.66	85.27	84.92	85.62
All	1400	64.49	109.37	86.03	89.03	83.23

Cuadro 4.1: Results of the training NER models.

From the table 4.1, we can draw several conclusions about the training of the Named Entity Recognition (NER) models in different experiments:

- The Basque experiment, despite a relatively high Loss Transformer and Loss NER, achieved the best F-score, Recall, and Precision metrics, implying that the model trained on the Basque language data was able to effectively recognize and correctly predict the entities from the text.
- The experiment involving both Spanish and Italian languages presented balanced results across F-score, Recall, and Precision, which are relatively high, suggesting that combining these languages did not significantly impact the model's performance.
- The experiment with the highest number of steps, Basque-Italian, does not necessarily exhibit the highest F-score, Recall, or Precision, which points to the

Evaluation

fact that simply increasing the number of training steps doesn't always lead to better performance.

- When all languages are included (All), the model demonstrates a high F-score, Recall, and Precision, suggesting that a multilingual approach can still deliver a commendable performance.

Experiment	Steps	Loss Transf.	Loss RE	F-score	Recall	Precision
Spanish	500	0.02	2.39	89.13	90.11	88.17
Italian	800	0.01	1.48	75.71	67.09	86.89
Basque	400	0.11	1.94	77.70	63.53	100.00
Spanish-Italian	800	0.06	1.67	81.62	69.38	99.11
Spanish-Basque	300	0.55	2.36	79.26	66.05	99.07
Basque-Italian	400	0.24	2.04	75.56	60.71	100.00
All	500	0.20	1.92	80.84	70.61	94.54

Cuadro 4.2: Results of the training RE models.

The table 4.2 presents the performance of the RE models across several experiments. Here are the conclusions that can be drawn from this data:

- The Spanish experiment stands out with the highest F-score and Recall, and quite high Precision. Interestingly, it managed to achieve these high scores with relatively more training steps and minimal losses. This indicates that the RE model trained with Spanish data has shown remarkable effectiveness in predicting relationships between entities in the text.
- Both the Basque and Basque-Italian experiments achieved perfect Precision (100.00), despite relatively lower Recall scores and higher Loss RE values. This means that while these models may not identify all true relationships (lower Recall) when they do make a prediction, it is likely to be correct (high Precision). This could potentially be attributed to the lack of sample variance presented to the model, resulting in imbalanced data. Additionally, the samples within these experiments might be too similar to each other, further contributing to this observed performance.
- The Italian and Basque-Italian experiments, which had similar amounts of training steps, differed quite noticeably in their performance metrics. This suggests that combining languages can have a complex effect on the performance of the RE models, and that this effect is not simply additive or predictable.
- Across all experiments, there is no clear correlation between the number of training steps and the model's performance metrics. For instance, the Spanish-Basque experiment with the fewest training steps (300) did not lead to the worst nor the best performance. This supports the idea that optimal training length can depend on many factors, including the specific language or combination of languages used.
- When considering all languages, the F-score, Recall, and Precision metrics are relatively balanced and moderately high, suggesting that the model has a decent overall performance in identifying relations across languages. However, it is not at the top in any single metric, indicating potential room for optimization.

Experiment	F-score	Recall	Precision
Spanish	86.49	87.27	85.71
Italian	79.19	70.24	90.77
Basque	75.16	61.11	97.58
Spanish-Italian	86.65	81.78	92.14
Spanish-Basque	82.96	73.14	95.82
Basque-Italian	75.50	61.51	97.71
All	81.87	79.01	84.95

Cuadro 4.3: Evaluation results of the trained RE models.

The table 4.3 presents the evaluation performance on the threshold 0.5 of the RE models the experiments. The threshold of 0.5 was chosen because it is a common practice in binary classification tasks, such as predicting the presence or absence of a relation. With the model outputting a continuous score between 0 and 1, the threshold serves as the decision boundary. Values equal to or above the threshold are classified as 'True' for the relation (REL), while those below are classified as 'False'. A threshold of 0.5 is the natural choice when there is no initial bias towards either class and the cost of false positives and false negatives is considered to be roughly the same. It essentially means that if the model is at least 50% sure that a relation exists, it will predict it as 'True':

- Spanish and Spanish-Italian experiments have shown the most balanced results among all tested scenarios, with relatively high F-scores, recalls, and precisions, indicating a good overall performance.
- On the other hand, models trained on Italian, Basque, and Basque-Italian experiments seem to have a high bias towards precision, reflected in the values above 90%. However, the recall for these experiments is noticeably lower, resulting in reduced F-scores. This indicates that these models might be highly accurate in predicting positive instances but fail to capture all true positive instances, suggesting a need for improved recall.
- The Spanish-Basque experiment also shows high precision, albeit with a slightly improved recall compared to Italian and Basque. However, its performance is still below Spanish and Spanish-Italian.
- The All experiment, which presumably combines all the data, shows a good balance between precision and recall. Its F-score is lower compared to the Spanish and Spanish-Italian experiments but higher than the rest, indicating an overall good but not optimal performance.

The performance varies considerably across different experiments, demonstrating the importance of language context in relationship extraction and named entity recognition tasks. Conclusions are mainly:

- **Spanish and Spanish-Italian experiments:** These models showcase an impressive balance between precision, recall, and F-score, with Spanish leading in terms of F-score and recall across all experiments. This suggests that the models trained on Spanish, or a mixture of Spanish and Italian data, are most effective in predicting relations in the dataset.

- **Italian, Basque, and Basque-Italian experiments:** Despite the strong performance of the basque model for NER, this model with Italian and Basque-Italian display high precision but lower recall. Consequently, while the predicted relations are likely correct, many actual relations are not detected, impacting the F-score negatively. Efforts to improve these models should focus on enhancing recall without compromising precision.
- **Spanish-Basque experiments:** The results are similar to those of the Italian, Basque, and Basque-Italian experiments, with relatively high precision but lower recall and F-score. As with the previous group, these models need improvement in their ability to identify all actual relations.
- **All experiment:** The experiment that utilizes all languages in training shows moderate, well-rounded results in all three metrics among all models. However, the performance does not excel in any specific area, indicating that further tuning and optimization are necessary to improve its effectiveness.

In conclusion, the choice of language data greatly influences the models' performance. Furthermore, the models tend to be more accurate (high precision) than comprehensive (high recall), with a tendency to miss existing relationships. Future work should address this disparity to improve the models' utility and reliability in diverse linguistic contexts.

4.3. Discussion

The dominance of English-oriented models can have effects on the performance of AI systems when applied to other languages, such as Spanish.

It is important to recognize the potential for bias in foundational models, such as those developed by Hugging Face, where English is often the primary language of base models.

This bias can lead to less-than-optimal performance when these models are applied to other languages, which is especially critical in global fields like healthcare that significantly impact people's lives.

While utilizing multilingual models offers a partial solution, it may still fall short of providing the level of language-specific fine-tuning needed for optimal performance in certain tasks.

Moreover, the community around this field could advance multilingual foundational models, specifically by cultivating richer, high-quality datasets in several languages. This would enable more accurate solutions, capable of supporting a wide range of languages, thus reducing the dependency on English models.

One significant area is the potential applications of the work, extending beyond healthcare into other sectors where linguistic diversity is key.

For instance, these models could find uses in other fields such as international law enforcement, multinational corporations, or global non-profit organizations, among others.

With increasing computational power and sophisticated modelling techniques, there is the potential to scale these models to get more languages and larger datasets. This

can make these NER and RE tools more robust and widely applicable with the ability to handle increasingly complex linguistic structures and contexts.

The continually advancements in AI could lead to more efficient and accurate models, improved integration with other systems, and the development of new applications.

The role of all these technologies might significantly influence the future impact and relevance of this research.

Capítulo 5

Conclusions and Future Work

In conclusion, it is evident that the choice of language data plays a crucial role in shaping the performance of models. The data used to train these models directly impacts their ability to understand and generate language. Therefore, careful consideration must be given to the selection and quality of the language data utilized.

Furthermore, it is important to note that these models typically prioritize accuracy, often referred to as high precision, over comprehensiveness, or high recall. This means that while the models excel at providing accurate responses, they may sometimes miss out on capturing all the existing relationships within the data.

Addressing this disparity between precision and recall is a critical area for future research. Improving the models' utility and reliability in diverse linguistic contexts requires efforts to enhance their ability to recognize and comprehend complex relationships. This could involve exploring new techniques to optimize recall without sacrificing precision, allowing the models to capture a more comprehensive understanding of language.

By addressing these challenges, future work can contribute to the development of language models that are not only accurate but also possess a broader understanding of the intricate nuances and connections present in natural language. Ultimately, this will lead to more reliable and useful models, capable of effectively processing and generating language in a wide range of linguistic contexts.

Other possible future work for this research could be divided into several ways. One primary direction could be the extension of this research to other languages and regions. The current work is focusing on Spanish, Italian, and Basque, so it could be expanded to include other languages.

Investigation into improving the integration between transformer-based models and SpaCy's functionalities could be another promising area for future work. This might involve developing more sophisticated pipelines or finding more efficient ways to combine these technologies. This work could be open by the companies involved to let the community contribute on it.

Future work could explore the application of these NER and RE models in various domains beyond healthcare. While the focus of the current work is on medical data, the methodologies and strategies developed could be adapted to other areas such as legal documents, scientific literature, news articles, and more.

On the other hand, the use of multimodal models which mix information from different data types or modes, such as text, image, and even audio. These models can potentially provide a richer understanding of the data by leveraging the complementary information from each modality. For example, in the biomedical field, a multimodal model could leverage both text descriptions and associated images to more accurately identify and understand the relationship between biological entities.

LayoutXLM [29] is a multilingual expansion of LayoutLMv2 that incorporates image information during pre-training, enabling visual and textual interaction. It outperforms existing solutions in multilingual document understanding, bridging language gaps. LiLT [30], on the other hand, overcomes language limitations by pre-training structured documents in one language and adapting to others, resulting in competitive performance regardless of the pre-training language. However, a study comparing LayoutXLM and a graph neural network (GNN) for document relation extraction [31] found that GNN outperforms LayoutXLM in this specific task. While both models achieve similar results, they differ in how they integrate multiple modalities, with LayoutXLM fusing them during pre-training and GNN cascading them.

The progress in knowledge graph techniques in the medical sector is fostering the creation of increasingly sophisticated and useful intelligent applications. A Knowledge Graph, a structured information network with entities represented as nodes and their relations as edges, facilitates complex semantic relationships and hierarchies. Its utility spans areas like AI, search engines, and natural language processing, owing to its proficiency in data integration, querying, and analysis. These advancements are beneficial to a range of individuals in the healthcare field, including clinicians, students, patients, researchers, and administrative staff. The ultimate goal is to develop an artificial intelligence system inspired by human cognition, capable of integrating generic information, common sense knowledge and domain-specific knowledge, together with social norms and values and individual cognitive models.

The papers hereafter highlight the importance of medical knowledge graphs (MKGs) in healthcare applications. [32] explore MKG formation, addressing challenges such as managing heterogeneous data and ensuring accurate knowledge representation. [33] propose a multimodal approach for COVID-19 diagnosis, combining medical images and doctor-patient dialogues to create a knowledge graph, resulting in effective disease detection. [34] focus on classifying Diagnosis Related Groups (DRGs) in Chinese medicine, utilizing a dynamic knowledge graph and demonstrating the effectiveness of integrating domain knowledge.

Overall, these studies emphasize the significance of MKGs in improving healthcare systems by addressing data complexity, incorporating multimodal information, and leveraging domain-specific knowledge for diagnosis and classification tasks.

So finally, it is feasible to construct a knowledge graph based on the outputs from the NER and RE models applied to user inputs. Once this knowledge graph is established, it can be compared and contrasted with more comprehensive knowledge graphs. This comparison would allow us to determine what information has been extracted from the user input and what additional knowledge can be added from potential ontologies. This would serve to make a recommendation system inside of the medical platform to help doctors to visualize the context presented by the patient.

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Appendix

Epoch	Steps	Loss Transformer	Loss NER	F-score	Recall	Precision	Score
0	0	72.84	160.03	0.81	1.23	0.61	0.01
8	200	3213.67	7691.08	66.03	70.07	62.42	0.66
16	400	160.84	309.02	74.43	79.86	69.70	0.74
24	600	22.01	43.20	76.43	80.54	72.73	0.76
32	800	0.00	0.00	76.68	81.08	72.73	0.77
40	1000	0.00	0.00	76.53	81.51	72.12	0.77

Cuadro 1: Training process **Spanish NER**.

Epoch	Steps	Loss Transformer	Loss RE	F-score	Recall	Precision	Score
0	0	0.07	0.35	21.20	98.90	11.87	0.21
14	100	15.35	9.86	82.58	70.33	100.00	0.83
29	200	0.31	3.23	87.50	84.62	90.59	0.87
44	300	0.22	2.89	87.50	92.31	83.17	0.87
59	400	0.06	2.49	89.01	89.01	89.01	0.89
74	500	0.02	2.39	89.13	90.11	88.17	0.89
88	600	0.02	2.33	89.01	89.01	89.01	0.89
103	700	0.00	2.38	87.95	80.22	97.33	0.88

Cuadro 2: Training process **Spanish RE**.

Threshold	F-score	Recall	Precision
0.00	15.21	100.00	8.23
0.05	15.41	96.97	8.37
0.10	15.54	92.73	8.48
0.20	15.26	82.42	8.41
0.30	14.47	69.70	8.08
0.40	13.47	55.76	7.66
0.50	13.22	47.88	7.67
0.60	12.81	38.79	7.67
0.70	12.90	30.91	8.15
0.80	11.62	20.00	8.19
0.90	7.85	9.09	6.91
0.99	1.08	0.61	4.76
1.00	0.00	0.00	0.00

Cuadro 3: Random baseline **Spanish RE** evaluation results.

Threshold	F-score	Recall	Precision
0.00	15.21	100.00	8.23
0.05	75.25	92.12	63.60
0.10	81.50	92.12	73.08
0.20	84.21	92.12	77.55
0.30	84.59	91.52	78.65
0.40	85.88	90.30	81.87
0.50	86.49	87.27	85.71
0.60	86.49	87.27	85.71
0.70	84.85	76.36	95.45
0.80	84.35	75.15	96.12
0.90	83.96	74.55	96.09
0.99	64.54	49.09	94.19
1.00	28.28	16.97	84.85

Cuadro 4: Trained **Spanish RE** evaluation results.

Epoch	Steps	Loss Transformer	Loss NER	F-score	Recall	Precision	Score
0	0	951.81	91.17	0.00	0.00	0.00	0.00
7	200	46613.40	11060.36	54.80	68.42	45.70	0.55
14	400	833.72	1768.76	75.00	75.00	75.00	0.75
21	600	174.80	314.87	75.30	76.83	73.83	0.75
28	800	207.76	316.13	74.82	70.21	80.08	0.75
35	1000	123.23	124.82	76.92	75.76	78.12	0.77
42	1200	42.13	52.17	77.52	76.92	78.12	0.78

Cuadro 5: Training process **Italian NER**.

Epoch	Steps	Loss Transformer	Loss RE	F-score	Recall	Precision	Score
0	0	0.08	0.10	0.00	0.00	0.00	0.00
13	100	7.15	5.39	59.13	43.04	94.44	0.59
26	200	1.09	2.43	68.75	55.70	89.80	0.69
40	300	0.35	2.08	71.64	60.76	87.27	0.72
53	400	0.09	1.70	73.60	58.23	100.00	0.74
66	500	0.05	1.56	75.36	65.82	88.14	0.75
79	600	0.03	1.50	75.52	68.35	84.38	0.76
92	700	0.02	1.59	76.06	68.35	85.71	0.76
105	800	0.01	1.48	75.71	67.09	86.89	0.76

Cuadro 6: Training process **Italian RE**.

Threshold	F-score	Recall	Precision
0.00	9.59	100.00	5.04
0.05	9.83	97.62	5.17
0.10	9.68	91.67	5.11
0.20	9.56	82.14	5.07
0.30	9.26	69.05	4.96
0.40	9.80	63.10	5.31
0.50	9.97	54.76	5.48
0.60	10.13	45.24	5.71
0.70	11.01	36.90	6.47
0.80	10.40	25.00	6.56
0.90	11.52	16.67	8.81
0.99	2.11	1.19	9.09
1.00	0.00	0.00	0.00

Cuadro 7: Random baseline **Italian RE** evaluation results.

Threshold	F-score	Recall	Precision
0.00	9.59	100.00	5.04
0.05	78.15	70.24	88.06
0.10	78.67	70.24	89.39
0.20	78.67	70.24	89.39
0.30	78.67	70.24	89.39
0.40	78.67	70.24	89.39
0.50	79.19	70.24	90.77
0.60	76.81	63.10	98.15
0.70	76.81	63.10	98.15
0.80	76.81	63.10	98.15
0.90	76.81	63.10	98.15
0.99	73.13	58.33	98.00
1.00	55.93	39.29	97.06

Cuadro 8: Trained **Italian RE** evaluation results.

Epoch	Steps	Loss Transformer	Loss NER	F-score	Recall	Precision	Score
0	0	107.60	92.99	1.85	2.69	1.41	0.02
6	200	5634.09	11227.28	78.66	85.43	72.88	0.79
12	400	655.01	1492.58	86.89	90.77	83.33	0.87
18	600	147.43	325.25	87.54	89.88	85.31	0.88
24	800	65.04	125.54	89.58	89.33	89.83	0.90
31	1000	39.72	70.26	88.33	87.96	88.70	0.88
37	1200	18.80	20.68	88.79	89.17	88.42	0.89
43	1400	10.20	11.31	88.95	89.20	88.70	0.89
50	1600	1.57	1.53	88.57	89.60	87.57	0.89
56	1800	0.16	0.30	88.60	89.37	87.85	0.89

Cuadro 9: Training process **Basque NER**.

Epoch	Steps	Loss Transformer	Loss RE	F-score	Recall	Precision	Score
0	0	0.02	0.09	0.00	0.00	0.00	0.00
11	100	10.78	5.88	67.63	55.29	87.04	0.68
24	200	1.17	2.66	70.15	55.29	95.92	0.70
36	300	0.33	2.16	77.14	63.53	98.18	0.77
48	400	0.11	1.94	77.70	63.53	100.00	0.78

Cuadro 10: Training process **Basque RE**.

Threshold	F-score	Recall	Precision
0.00	7.98	100.00	4.16
0.05	8.11	96.97	4.23
0.10	8.04	91.41	4.21
0.20	8.08	81.82	4.25
0.30	8.46	75.76	4.48
0.40	8.22	63.64	4.39
0.50	8.38	55.05	4.54
0.60	8.61	45.96	4.75
0.70	8.40	34.85	4.78
0.80	8.16	24.24	4.90
0.90	7.14	12.63	4.98
0.99	3.40	2.02	10.81
1.00	0.00	0.00	0.00

Cuadro 11: Random baseline **Basque RE** evaluation results.

Threshold	F-score	Recall	Precision
0.00	7.98	100.00	4.16
0.05	75.00	63.64	91.30
0.10	75.90	63.64	94.03
0.20	75.99	63.13	95.42
0.30	75.84	62.63	96.12
0.40	75.16	61.11	97.58
0.50	75.16	61.11	97.58
0.60	75.00	60.61	98.36
0.70	75.00	60.61	98.36
0.80	73.82	59.09	98.32
0.90	73.25	58.08	99.14
0.99	63.70	46.97	98.94
1.00	42.23	26.77	100.00

Cuadro 12: Trained **Basque RE** evaluation results.

Epoch	Steps	Loss Transformer	Loss NER	F-score	Recall	Precision	Score
0	0	361.32	94.65	0.00	0.00	0.00	0.00
3	200	86328.65	11398.28	55.43	52.92	58.19	0.55
7	400	1327.37	2793.07	77.66	79.90	75.53	0.78
11	600	330.66	748.53	78.49	75.72	81.47	0.78
15	800	130.87	295.40	81.09	80.71	81.47	0.81
19	1000	68.54	116.37	81.09	81.19	81.00	0.81
23	1200	44.07	70.81	82.00	84.04	80.05	0.82
27	1400	59.08	68.71	81.57	81.67	81.47	0.82

Cuadro 13: Training process **Spanish-Italian NER**.

Epoch	Steps	Loss Transformer	Loss RE	F-score	Recall	Precision	Score
0	0	0.05	0.09	0.00	0.00	0.00	0.00
7	100	9.54	6.23	65.40	53.75	83.50	0.65
14	200	2.29	3.18	69.40	68.75	70.06	0.69
21	300	0.51	2.60	78.62	71.25	87.69	0.79
28	400	0.23	2.14	81.82	73.12	92.86	0.82
36	500	0.17	1.89	82.35	74.38	92.25	0.82
43	600	0.09	1.69	82.52	73.75	93.65	0.83
50	700	0.09	1.68	82.93	74.38	93.70	0.83
57	800	0.06	1.67	81.62	69.38	99.11	0.82
65	900	0.03	1.54	82.98	73.12	95.90	0.83
72	1000	0.02	1.61	82.05	70.00	99.12	0.82

Cuadro 14: Training process **Spanish-Italian RE**.

Threshold	F-score	Recall	Precision
0.00	13.13	100.00	7.03
0.05	12.93	93.80	6.94
0.10	12.84	89.15	6.92
0.20	12.26	76.36	6.66
0.30	12.09	67.05	6.64
0.40	11.37	54.65	6.34
0.50	11.10	45.35	6.32
0.60	10.82	36.05	6.37
0.70	11.25	28.68	6.99
0.80	10.10	18.99	6.88
0.90	6.57	7.75	5.70
0.99	0.69	0.39	3.23
1.00	0.00	0.00	0.00

Cuadro 15: Random **Spanish-Italian RE** evaluation results.

Threshold	F-score	Recall	Precision
0.00	13.13	100.00	7.03
0.05	85.66	84.50	86.85
0.10	87.20	84.50	90.08
0.20	87.20	84.50	90.08
0.30	87.20	84.50	90.08
0.40	87.15	84.11	90.42
0.50	86.65	81.78	92.14
0.60	86.74	79.84	94.93
0.70	86.44	79.07	95.33
0.80	86.44	79.07	95.33
0.90	86.44	79.07	95.33
0.99	80.09	69.38	94.71
1.00	55.91	40.31	91.23

Cuadro 16: Trained **Spanish-Italian RE** evaluation results.

Epoch	Steps	Loss Transformer	Loss NER	F-score	Recall	Precision	Score
0	0	146.00	94.17	1.24	0.96	1.73	0.01
3	200	5959.05	11399.25	69.40	83.04	59.61	0.69
7	400	904.11	2254.73	85.90	87.48	84.39	0.86
10	600	281.33	675.07	86.34	84.01	88.80	0.86
14	800	137.90	284.14	87.31	90.17	84.63	0.87
18	1000	72.05	150.18	87.55	89.40	85.77	0.88
21	1200	25.81	52.65	87.31	87.48	87.14	0.87
25	1400	12.72	22.09	86.99	86.32	87.67	0.87
29	1600	15.18	25.18	87.38	85.36	89.49	0.87
32	1800	4.31	6.51	87.71	87.28	88.13	0.88

Cuadro 17: Training process **Spanish-Basque NER**.

Epoch	Steps	Loss Transformer	Loss RE	F-score	Recall	Precision	Score
0	0	0.05	0.08	0.00	0.00	0.00	0.00
6	100	11.37	5.82	65.15	53.09	84.31	0.65
13	200	1.28	2.74	77.62	68.52	89.52	0.78
20	300	0.55	2.36	79.26	66.05	99.07	0.79
27	400	0.29	2.12	80.29	67.90	98.21	0.80
34	500	0.10	2.04	80.59	67.90	99.10	0.81

Cuadro 18: Training process **Spanish-Basque RE**.

Threshold	F-score	Recall	Precision
0.00	10.53	100.00	5.56
0.05	10.37	93.62	5.49
0.10	10.27	88.03	5.45
0.20	10.39	80.05	5.56
0.30	10.48	71.01	5.66
0.40	10.62	62.77	5.80
0.50	10.22	50.80	5.68
0.60	10.26	41.76	5.85
0.70	9.87	30.85	5.87
0.80	8.70	19.95	5.56
0.90	7.25	9.84	5.75
0.99	2.27	1.33	7.81
1.00	0.52	0.27	16.67

Cuadro 19: Random baseline **Spanish-Basque RE** evaluation.

Threshold	F-score	Recall	Precision
0.00	10.53	100.00	5.56
0.05	80.64	73.67	89.07
0.10	81.71	73.67	91.72
0.20	82.14	73.40	93.24
0.30	82.83	73.14	95.49
0.40	82.96	73.14	95.82
0.50	82.96	73.14	95.82
0.60	82.73	72.61	96.13
0.70	82.50	72.07	96.44
0.80	82.50	72.07	96.44
0.90	81.23	70.21	96.35
0.99	72.31	57.98	96.04
1.00	45.34	29.79	94.92

Cuadro 20: Trained **Spanish-Basque RE** evaluation.

Epoch	Steps	Loss Transformer	Loss NER	F-score	Recall	Precision	Score
0	0	270.87	100.53	0.00	0.00	0.00	0.00
3	200	29532.10	12574.22	76.94	80.00	74.10	0.77
6	400	1060.23	2721.38	81.45	74.39	90.00	0.81
10	600	415.49	910.63	84.82	81.54	88.36	0.85
13	800	219.64	457.04	83.14	88.97	78.03	0.83
17	1000	154.81	318.59	84.50	89.54	80.00	0.85
20	1200	79.95	134.37	84.70	85.48	83.93	0.85
24	1400	66.04	74.66	84.72	84.03	85.41	0.85
27	1600	37.60	48.83	85.00	83.65	86.39	0.85
31	1800	16.92	22.66	85.27	85.62	84.92	0.85
35	2000	16.39	27.98	85.20	85.48	84.92	0.85

Cuadro 21: Training process **Basque-Italian NER**.

Epoch	Steps	Loss Transformer	Loss RE	F-score	Recall	Precision	Score
0	0	0.04	0.09	0.00	0.00	0.00	0.00
6	100	10.75	5.78	59.26	47.62	78.43	0.59
13	200	2.24	2.96	60.33	43.45	98.65	0.60
20	300	0.67	2.42	74.29	61.90	92.86	0.74
26	400	0.24	2.04	75.56	60.71	100.00	0.76

Cuadro 22: Training process **Basque-Italian RE**.

Threshold	F-score	Recall	Precision
0.00	8.29	100.00	4.32
0.05	8.36	96.04	4.37
0.10	8.10	88.49	4.24
0.20	8.22	80.94	4.33
0.30	8.06	69.78	4.28
0.40	7.80	58.63	4.18
0.50	7.88	50.00	4.28
0.60	7.81	40.29	4.32
0.70	7.24	29.50	4.12
0.80	7.22	20.50	4.38
0.90	4.31	7.19	3.07
0.99	0.00	0.00	0.00
1.00	0.00	0.00	0.00

Cuadro 23: Random baseline **Basque-Italian RE** evaluation.

Threshold	F-score	Recall	Precision
0.00	8.29	100.00	4.32
0.05	73.93	62.23	91.05
0.10	74.57	62.23	93.01
0.20	75.22	62.23	95.05
0.30	75.16	61.51	96.61
0.40	75.33	61.51	97.16
0.50	75.50	61.51	97.71
0.60	75.28	60.79	98.83
0.70	75.28	60.79	98.83
0.80	75.17	60.43	99.41
0.90	74.04	58.99	99.39
0.99	56.19	39.21	99.09
1.00	12.12	6.47	94.74

Cuadro 24: Trained **Basque-Italian RE** evaluation.

Epoch	Steps	Loss Transformer	Loss NER	F-score	Recall	Precision	Score
0	0	3857.60	107.86	0.00	0.00	0.00	0.00
2	200	39562.10	12623.66	63.83	61.08	66.84	0.64
4	400	1493.47	3665.38	79.40	74.66	84.77	0.79
7	600	555.23	1260.31	84.09	88.46	80.13	0.84
9	800	287.71	660.15	83.42	85.81	81.16	0.83
12	1000	160.92	286.77	84.83	84.88	84.77	0.85
14	1200	91.34	159.09	84.96	86.50	83.48	0.85
16	1400	64.49	109.37	86.03	83.23	89.03	0.86
19	1600	27.18	43.49	85.90	84.61	87.23	0.86
21	1800	17.06	27.18	85.80	84.31	87.35	0.86
24	2000	21.58	29.34	86.28	85.35	87.23	0.86

Cuadro 25: Training process **All NER**.

Epoch	Steps	Loss Transformer	Loss RE	F-score	Recall	Precision	Score
0	0	0.10	0.10	0.00	0.00	0.00	0.00
7	100	8.62	5.66	63.66	54.69	76.14	0.64
14	200	1.76	2.88	68.98	56.73	87.97	0.69
21	300	0.63	2.53	68.10	51.84	99.22	0.68
28	400	0.26	2.15	78.70	69.39	90.91	0.79
35	500	0.20	1.92	80.84	70.61	94.54	0.81

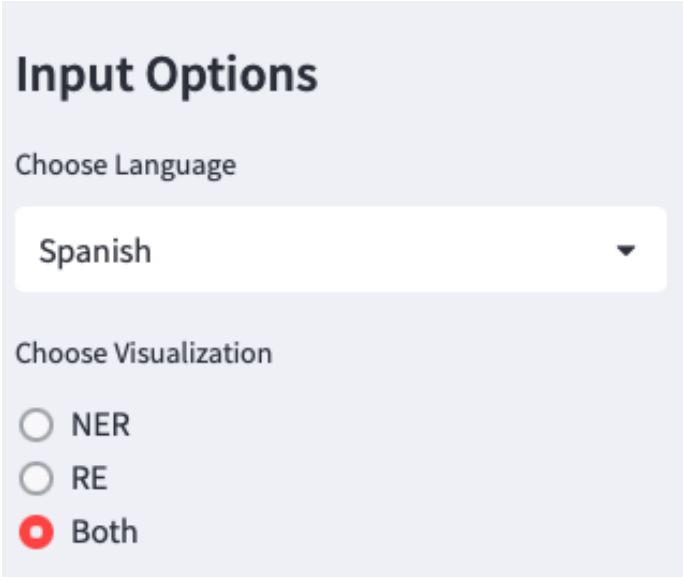
Cuadro 26: Training process **All RE**.

Threshold	F-score	Recall	Precision
0.00	11.94	100.00	6.35
0.05	11.76	94.46	6.27
0.10	11.70	89.50	6.26
0.20	11.58	79.01	6.25
0.30	11.46	69.10	6.25
0.40	11.67	60.93	6.45
0.50	11.06	49.27	6.23
0.60	10.75	39.36	6.22
0.70	10.46	29.74	6.34
0.80	9.04	18.66	5.96
0.90	7.25	9.33	5.93
0.99	1.52	0.87	5.88
1.00	0.00	0.00	0.00

Cuadro 27: Random baseline **All RE** evaluation.

Threshold	F-score	Recall	Precision
0.00	11.94	100.00	6.35
0.05	75.24	81.05	70.20
0.10	78.81	80.76	76.94
0.20	81.11	80.76	81.47
0.30	81.72	80.17	83.33
0.40	81.74	79.59	84.00
0.50	81.87	79.01	84.95
0.60	82.44	78.72	86.54
0.70	82.94	77.26	89.53
0.80	83.17	76.38	91.29
0.90	83.01	74.05	94.42
0.99	74.11	60.93	94.57
1.00	56.62	40.52	93.92

Cuadro 28: Trained **All RE** evaluation.



Input Options

Choose Language

Spanish ▼

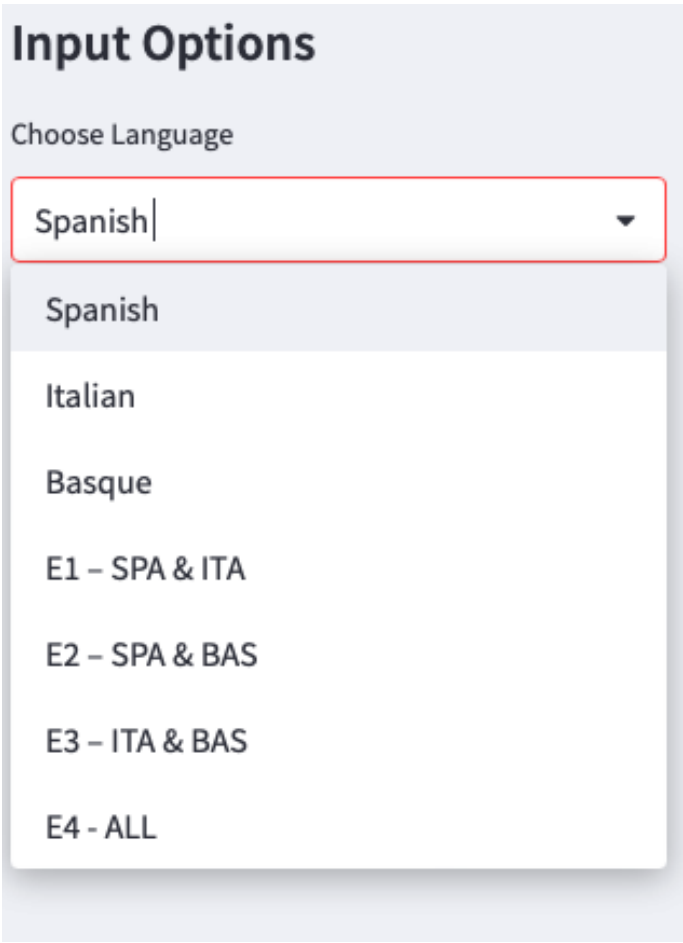
Choose Visualization

☐ NER

☐ RE

☒ Both

Figura 1: Buttons to choose which results are going to be visualized.



Input Options

Choose Language

Spanish ▼

Spanish

Italian

Basque

E1 – SPA & ITA

E2 – SPA & BAS

E3 – ITA & BAS

E4 - ALL

Figura 2: Sidebar to choose the experiment to be evaluated.

Paquita – A Named Entity Recognition and Relation Extraction Tool

Welcome to my NER and RE tool. Here you can input text and extract named entities as well as their relations.

Enter your text here

Paciente de 65 a. de edad, que presentaba una elevación progresiva de las cifras de PSA desde 6 ng/ml a 12 ng/ml en el último año. Dicho paciente había sido sometido un año antes a una biopsia transrectal de próstata ecodirigida por sextantes que fue negativa. El estudio espectroscópico de ésta lesión mostró una curva de colina discretamente más elevada que la curva de citrato, con un índice de Ch-Cr/Ci > 0,80, que sugería la presencia de lesión neoplásica, por lo que se biopsia dicha zona por ecografía transrectal. La AP de la biopsia confirmó la presencia de un ADK próstata Gleason 6.



Extract

Figura 3: Introducing input into the interface.

Named Entity Recognition Results

Paciente de 65 a. de edad, que presentaba una elevación progresiva de las cifras de **PSA** desde **6 ng/ml** a **12 ng/ml** en el último año. Dicho paciente había sido sometido un año antes a una biopsia transrectal de próstata ecodirigida por sextantes que fue **negativa**. El **estudio** espectroscópico de ésta lesión mostró una curva de colina discretamente más elevada que la curva de citrato, con un **índice** de Ch-Cr/ **Ci** **> 0,80,** que sugería la presencia de lesión neoplásica, por lo que se biopsia dicha zona por ecografía transrectal. La AP de la biopsia confirmó la presencia de un **ADK** próstata **Gleason** **6.**

Figura 4: NER visualization.

Relation Extraction Results

	Entity 1	Entity 2	Relationship
0	estudio	índice	0.3904532194137573
1	estudio	Ci	0.8297896981239319
2	estudio	> 0,80,	0.9941149353981018
3	índice	Ci	0.31696343421936035
4	índice	> 0,80,	0.9414492845535278
5	Ci	> 0,80,	0.4771077632904053

Figura 5: RE visualization.

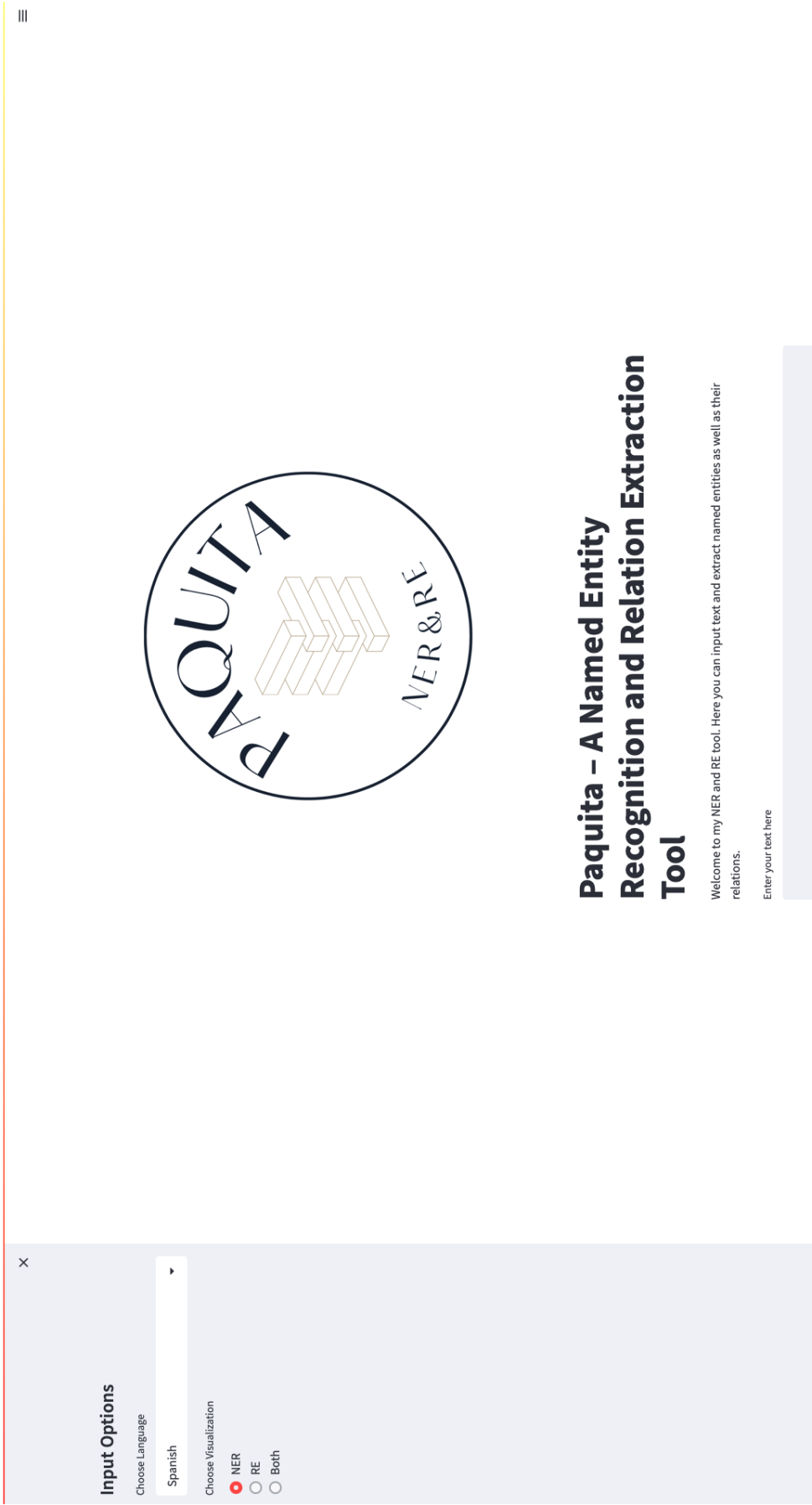


Figura 6: Interface visualization.